

Endogenous Belief Hazards

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Abstract

When an agent selects among models by testing rather than by averaging—holding one until its own record rejects it, then re-selecting—the rate at which she abandons it is an *endogenous belief hazard*, derived from the rule and governed by a state the belief does not contain. The primitive is Ortoleva’s, cast in continuous time. The belief is not a sufficient statistic for its own revision: the missing coordinate is the record under which it has survived, so two forecasters sharing every public-data posterior can face different hazards. Across a population that watches one path and differs only in surprise tolerance, the belief cross-section is the exact pushforward of that tolerance distribution. And because the trigger is the improbability of the record, not the payoff, the hazard is invariant to the stakes—the model’s structural restriction: what the data could refute, and what separates inference-governed revision from its cost-based alternatives.

Keywords: belief revision, hypothesis testing, non-Bayesian updating, adjustment hazard, sufficient statistic, heterogeneous beliefs.

JEL classification: D81, D83, D84, C12.

1 Introduction

Macroeconomics and finance infer beliefs from summaries, such as where they sit—the consensus forecast—and how spread out they are—the dispersion. Two forecasters who report the same numbers are, by those summaries, in the same state.¹ This paper is about a coordinate they leave out. Two forecasters can report three percent inflation and be very differently exposed to abandoning it. One reached the number only recently; her test record, reset when she adopted the model, is short and sits well inside what it tolerates—she is *robust*, and the surprise that made her drop a higher number chose her model without aging the record she now monitors. The other has held three percent through a long stretch in which the data crept toward the edge of her model’s ordinary range; her record, aged and near its limit, leaves her one surprise from revising—she is *fragile*. They forecast alike and revise unlike. What the held forecast misses is a belief’s *fragility*—the *hazard* of revision—carried not by tenure alone or by the surprise that first installed the model, but jointly by the *since-adoption* record under which it has survived (the data seen since adoption, the test record having reset then) and by its age, through the age-dependent band the record is tested against.

The behavior that makes the record matter is the way a cautious statistician—or an investor who acts like one—treats a model she is unsure of. Rather than average the candidate models into a posterior and act on the mixture, she holds the single model she finds most plausible and acts on it: she keeps it while the data look ordinary under it, and abandons it the first time the data look too unlikely to be a fluke, whereupon she reconsiders and adopts whatever model now looks best. This is the hypothesis-

¹Treating the belief as the state has a decision-theoretic warrant: Epstein and Le Breton (1993) show that dynamic consistency forces Bayesian updating, under which the belief is its own sufficient statistic. The hypothesis-testing agent here is not Bayesian, so that warrant lapses—and the coordinate the summaries omit is the one this paper recovers (§2.2).

testing rule Ortoleva (2012) axiomatizes, cast here in continuous time and closest in spirit to the model-validation learner of Cho and Kasa (2015). Beliefs under it do not drift a little with each number; they sit still, then jump. The object we study is the rate at which they jump: the *endogenous belief hazard*, the history-dependent rate of model abandonment the rule induces. It is derived from the rule, not posited, and it is the surprise-governed twin of the cost-governed adjustment hazard of the lumpy-adjustment literature (Caballero and Engel, 1999; Alvarez et al., 2022).

The primitives are spare. The agent carries a finite menu of candidate models and a ranking over it; she holds the model the ranking puts first and monitors the one statistic the menu disputes—an average growth rate when the models disagree about a trend, a count of disasters when they disagree about a crash rate. She keeps the held model while that statistic stays inside the *ordinary range* her model assigns probability at least $1 - \alpha$ —exactly $1 - \alpha$ in the continuous channel—a fixed-size acceptance band, and abandons it the first time the band is breached—whereupon she re-ranks the menu on the record just ended and commits to the new front-runner. The test size α is the only *behavioral* parameter—no adjustment cost, no signal noise, no bias—the menu, prior, and recording environment being the structural primitives any such rule fixes. Nor is the band a further choice: interval acceptance is the general form of an exact test of the held model, any other region being weakly dominated (Lemma A.2); and among size- α regions the primitive pins the selection—the continuous-time limit of Ortoleva’s one-shot test is the highest-density region, uniquely so when the held density has no flat plateau at the cutoff level (Lemma A.3; assembled, with its regularity conditions, as Proposition 3). The hazard is then a sharp object: we characterize its law and tail analytically rather than simulate, and follow it from the single forecaster up to the market.

The central result is that the record under which a belief has survived—its content

and its age—is an irreducible state variable. The record carries the hazard—a first-exit object, not a stochastic intensity, made precise in §2.3—so the held belief is *not* a sufficient statistic for its own revision, and neither is any posterior a Bayesian forms by updating a fixed prior on the public data. Two forecasters who have watched the same history share that posterior yet differ in fragility, because they adopted their models on different dates. The state is thereby *enlarged*: from the held belief to the belief together with the record under which it has survived, the revision hazard determined by that enlarged state rather than by the belief alone (Theorem 1). Everything else in the paper is what this enlargement produces—that a correct model is revised forever, that the cross-section is a pushforward of one history, and that revision is invariant to the stakes—taken in turn, first within the single forecaster, then across a population. Recurrence drives the dynamics, and it is not ours: even a correct model, tested at a fixed size and monitored without end, is eventually carried out of band by its own fluctuations, then abandoned and re-adopted—the statistician’s *sampling to a foregone conclusion* (Anscombe, 1954; Robbins, 1970), which we import (Proposition 1). What settles is the model’s *location* (the truth, or the nearest menu model); what never settles is its *revision*, so the surviving record—and with it fragility—stays alive even after the reported model has consolidated.

Across a population the individual rule becomes macroeconomics, and poses the question this paper answers: when forecasters share the data but not their patience, how does a common shock move the cross-section of their beliefs? A patient forecaster sits through news that tips an impatient one, so a common shock crosses the thresholds only of whoever was already close, and the consensus moves through them alone. The disagreement is not about *information*—every forecaster has seen the same history—but about who is still willing to live with it: the spread of beliefs is the spread of patience carried forward along the one path the economy has taken, with noth-

ing idiosyncratic to average away. The cross-section is pinned down exactly as the fixed distribution of thresholds seen through the common history (Proposition 2), and every aggregate is a functional of that history, with no market average to solve for—an aggregate revision set at the boundary, by the mass that crosses and where its re-selection sends it, never by an interior average.

Carried into specific environments, the same primitive delivers a cluster of documented patterns that standard models reach only by adding parameters—a complacent market that moves sharply when the mass it has left poised at the boundary is surprised (Caballero and Engel, 1999; Alvarez et al., 2022), lumpy individual forecasts against a sluggish consensus (Coibion and Gorodnichenko, 2015; Bordalo et al., 2020), disagreement that widens in crises (Mankiw et al., 2003), and experience effects (Malmendier and Nagel, 2011; Nagel and Xu, 2022), and, from the same decay, a multidecade *informational* trend on a stationary process (a belief overhang with a demographic half-life)—each with no private information and no *behavioral* parameter beyond α , heterogeneity entering through the tolerance law G —a primitive to be specified, not tuned—(developed in the Online Appendix). In the rare-disaster economy, the fear that spikes by surprise and recedes on schedule imprints, under any monotone kernel, a countercyclical price of jump risk—sign and timing from the primitive, the level conceded to the kernel. Each pattern is a statement about *observed*, model-changing revisions, conditional on the boundary-mass conditions of Section 3. The rule also says something sharp and testable about *why* people revise: the trigger lives in the record, not the payoffs, so the rate of revision does not depend on the stakes (Proposition 5). Raise what a forecaster stands to lose by being wrong—a trader’s position, a policymaker’s credibility—while leaving the data she sees untouched. An account in which revision is held off because acting is costly or attention scarce (Sims, 2003; Baley and Turén, 2025) predicts she now revises sooner; ours predicts she does

not move at all, because the test that triggers her sees only the record. The parting is a structural restriction—at a fixed inference state the stakes move revision under the cost account and not under ours—testable where stakes vary exogenously of that state, or once the state is recovered (Appendix F).

Related literature

The mechanism sits among several literatures without being contained by any. Its primitive—hold one model, abandon it when its own record rejects it—is Ortoleva’s hypothesis-testing rule, one of the non-Bayesian updating rules reviewed by Ortoleva (2024) and since extended within decision theory in directions orthogonal to ours (Cheng, 2022); the same primitive arose independently in macroeconomics as the *inferential expectations* of Menzies and Zizzo (2009), where an agent holds a belief until a fixed-size- α test rejects it and rational expectations is the $\alpha \rightarrow 1$ limit, lately extended to abrupt belief shifts (Henckel et al., 2025), with an empirical counterpart in the model-switching account of survey expectations (Branch, 2007). We hold the rule fixed and take it dynamic in continuous time. Where inferential expectations switches a representative agent’s false null *once* to the rational expectation, we test the model the agent *holds* against its own record at fixed size: a correct model is then rejected recurrently, and the belief cross-section becomes a pushforward of the tolerance distribution—neither available from a one-way, representative-agent switch. The fixed test size sits against a dynamic-consistency benchmark rather than carrying one: full dynamic consistency would force Bayesian updating (Epstein and Le Breton, 1993), so a rule that revises *by surprise* cannot have it; Weinstein (2017) shows that an α -spending rule bounding the *lifetime* probability of such non-Bayesian *paradigm shifts* keeps the agent approximately dynamically consistent—a guarantee the per-

test reading deliberately forgoes, retaining exactly the multiple-testing cost he flags, uncontrolled lifetime rejection of a valid model. Three comparisons carry the paper. The nearest dynamic relative is the validation learner of Cho and Kasa (2015), who runs a per-period specification test with a vanishing gain and a record-varying threshold, so that a correct model is spared in the limit and switches are rare. We sit at the opposite end of that family: the test size is *fixed*, so even a correct model is rejected recurrently, and it is that recurrence—not the presence of a better-fitting rival—that drives both the dynamics and the state enlargement. The trigger is correspondingly *absolute*—each model judged against its own implied band, not a rival’s fit—so a good record buys no immunity—each is the concern the relative-fit switchers hedge, inverted. The recurrence is the continuous-time, fixed-tolerance analogue of the escape dynamics of Cho et al. (2002), driven by the repeated-significance effect of fixed-size monitoring.

The nearest asset-pricing neighbor is the smooth-filtering account of Dew-Becker et al. (2025), where a constant true intensity is filtered into a continuously varying perceived one that delivers leverage, skewness, and volatility dynamics. We share the constant-intensity premise but generate perceived risk as a *saw-tooth*—an upward jump at each disaster that finds a complacent mass poised at the band edge, a scheduled decline between—from discrete abandonment, supplying the jump-and-timing surface smooth filtering does not and conceding the higher-moment dynamics to it. The divergence is structural before it is quantitative: their perceived intensity is a sufficient statistic in the public filtration, whereas ours turns on private adoption dates the public data do not encode (Theorem 1).

That last contrast is the broader theme. A large literature lets rare events and slow learning generate *persistent*, belief-driven dynamics (Wachter and Zhu, 2025; Kozlowski et al., 2020; Collin-Dufresne et al., 2016; Weitzman, 2007); in every case

the persistence is a property of the *belief*—the posterior is the state, and two agents who share it share their futures. Our enlargement is different in kind: the coordinate that governs revision is the record under which the held model has survived, which a fixed-prior posterior on the public filtration does not encode (Theorem 1), so two forecasters with the *same* belief can face different hazards. Nor is the distinction Bayesian versus non-Bayesian—even the misspecified filters Dew-Becker et al. (2025) allow are recursions in the belief, which stays sufficient for its own motion; what we add is a setting in which it is not.

A second contrast separates the mechanism from the extrapolative and diagnostic-expectations account of belief-driven dynamics (Bordalo et al., 2018, 2020), in which a systematic distortion—beliefs that overreact by a kernel of truth—makes predictable forecast errors and predictable outcomes the same object. Our agent carries no such distortion: the held model consolidates on the Kullback–Leibler closest menu model (Proposition 1)—the truth itself when the menu contains it—so any forecast bias is, past consolidation, the fixed projection gap, constant across the revision cycle rather than a state-dependent distortion. From that point the dissociation is structural: revision carries no forecast-error predictability of its own, which a kernel-of-truth model cannot produce, since there the two predictabilities are one distortion—and it is the sharpest line between this mechanism and the overreaction family.

2 The rule and the endogenous hazard

A belief’s fragility lives not in where it sits but in how its record stands against the model holding it up. To make that precise we need the rule that maps a record to a revision, and the one feature that sets it apart: revision is triggered by surprise, not by the cost of being wrong or a rival’s better fit. We state the rule in continuous

time, show that the surviving record it tests is an irreducible state coordinate—the held belief is not a sufficient statistic for its own revision—and derive the endogenous belief hazard that enlarged state governs. The technical groundwork, and the one modeling choice continuous time forces, are in Appendix A.

2.1 The rule in continuous time

Fix a probability space carrying the data, with $\mathbb{P}^*$ the physical (true) law and $(\mathcal{F}_t)_{t \geq 0}$ the filtration generated by the observed path. Consider an investor who watches the data continuously. We state her rule in three parts.

Setup. The data are a process $X = (X_t)_{t \geq 0}$. The investor entertains a finite menu of models $\Theta = \{\theta_1, \dots, \theta_K\}$, each a candidate law for X on $(\mathcal{F}_t)$, and a second-order belief $\rho \in \Delta(\Theta)$ ranking them. She holds the top-ranked model

$$\hat{\theta}_t = \arg \max_{\theta \in \Theta} \rho_t(\theta). \quad (1)$$

Ties are broken by a fixed priority order on Θ (retaining the incumbent when it is among the maximizers), so $\hat{\theta}_t$ is single-valued and $(\mathcal{F}_t)$ -adapted; under the continuous record law *data-induced* likelihood ties at fixed times are $\mathbb{P}^*$ -null—a carried tie, the allowed tied initial ranking frozen through a spell, persists by construction and is resolved by the order—a tied *initial* ranking is possible by assumption, and no almost-sure statement below turns on tie-nullness at the stopped re-selection times—the order, not nullness, keeps the selection single-valued there; in the counting channel ties have positive probability outright and the priority order is then substantive—part of the rule, keeping the selection deterministic and the held-model process adapted—and the results below hold for any fixed choice of that order. Let σ_t be the last time

she revised (or 0), so that

$$a_t := t - \sigma_t \tag{2}$$

is the age of her current model and $X_{(\sigma_t, t]}$ the data observed since she adopted it—her track record for the held model. We require the menu laws pairwise mutually locally equivalent on $(\mathcal{F}_t)$, so the record has a density under each model and the cross-menu likelihood ratios in (8) are well-defined (Appendix A).

(i) What she monitors. She judges her model on the one feature of the record the candidate models disagree about: the average growth rate, if they dispute the trend; the number of disasters, if they dispute the crash rate. Call this summary T_a —the minimal sufficient statistic of the record for the menu (Appendix A)—and write its held-model density at age a as f_a^θ . The record accrues at a fixed *recording resolution* $\delta > 0$; the continuous-time objects below are its $\delta \downarrow 0$ limit (Appendix A).

Assumption 1 (Exponential-Lévy menu). *The menu is an exponential family of Lévy processes: there is an additive sufficient statistic S_a —a Lévy functional of the record, with (T_a, a) a one-to-one function of (S_a, a) —the summary recovering S_a at each fixed age—such that for any two menu laws the record’s likelihood ratio through age a has the Wald form*

$$\left. \frac{dP^\theta}{dP^{\theta'}} \right|_{\mathcal{F}_a} = \exp((\eta_\theta - \eta_{\theta'}) S_a - a(\psi_\theta - \psi_{\theta'})), \tag{3}$$

a pathwise function of (S_a, a) holding for every age a , $\mathbb{P}$ -a.s., with η_θ the canonical parameter and ψ_θ the cumulant normalizer of model θ —only the differences $\psi_\theta - \psi_{\theta'}$ enter, so the normalizers are fixed up to one common shift, and the channel instances below take the natural gauge.

Both channels are instances: the disputed-trend menu on a Brownian-with-drift

record has $S_a = X_a$, $\eta_\theta = \mu_\theta/\sigma^2$, $\psi_\theta = \mu_\theta^2/(2\sigma^2)$, and $T_a = \bar{X}_a = S_a/a$ (Girsanov); the disputed-crash menu has $S_a = N_a$, $\eta_\theta = \log \lambda_\theta$, $\psi_\theta = \lambda_\theta$, and $T_a = N_a$. Two consequences are used below: (T_a, a) is minimal sufficient and Markov under each menu law, so the enlarged state $(\hat{\theta}, T_a, a)$ is *finite*-dimensional; and at any revision the stopped-record likelihood ratio is again (3), now at $a = \tau$. Exponential-Lévy is exactly the class in which a scalar statistic stays sufficient at stopping times, not merely fixed ages (Küchler and Sørensen, 1989; Küchler and Lauritzen, 1989; Küchler and Sørensen, 1998); outside it, the insufficiency (i) still holds with the raw since-adoption record—which carries its own length—in place of (T_a, a) , while the coordinate-wise irreducibility (ii) and the finite-dimensional closure (iii) are statements about the summary state and are stated at exponential-Lévy scope.

(ii) When she abandons it. She keeps her model as long as the observed summary looks ordinary under it, and abandons it the first time the summary looks too unlikely. Fix a tolerance $\alpha \in (0, 1)$ —the size of the retention test. The *ordinary range* is the set of outcomes her model finds most likely, carved out to carry probability $1 - \alpha$:

$$A_\alpha(a) = \{x : f_a^{\hat{\theta}}(x) \geq c_\alpha(a)\}, \quad c_\alpha(a) \text{ chosen so that } \mathbb{P}^{\hat{\theta}}(T_a \in A_\alpha(a)) \geq 1 - \alpha. \quad (4)$$

The region carries mass exactly $1 - \alpha$ in the continuous channel and is the smallest probability-ordered integer interval of at least that mass (the Sterne region) in the discrete one. It is the highest-density region: interval acceptance is the general form of an exact test (Lemma A.2), and among size- α regions the primitive pins this one—the continuous-time limit of Ortoleva’s one-shot test, uniquely so absent a flat plateau of the held density at the cutoff level (Lemma A.3; Appendix A)—taken closed, abandonment strictly outside (the continuous boundary is null, Lemma A.3).

Proposition 3 (Appendix A) is that chain assembled; Lemma D.7 (Appendix D.5) records which downstream objects survive any selection with the same asymptotic endpoints. A larger α tightens the band; “tolerance α ” is the type label throughout, decreasing in α .

She abandons her model at the first age, after a probation onset $a_0 \geq 0$, at which the summary leaves the ordinary range:

$$\tau = \inf\{a > a_0 : T_a \notin A_\alpha(a)\}. \quad (5)$$

The onset a_0 is a primitive of the recording environment, not a behavioral parameter: positive on a diffusive record, where monitoring from $a = 0$ would reject even the truth at arbitrarily small ages (the law of the iterated logarithm; Lemma A.7), and zero on a counting record, where the floor at zero rules out an early lower exit though a disaster can still trigger an upper one. The holding-time results of Section 2.4 rest on it. In the two channels we use throughout, the range is an explicit interval around what the model expects:

$$\text{disputed trend:} \quad A_\alpha(a) = \left\{ |\bar{X}_a - \hat{\mu}| \leq z_\alpha \sigma / \sqrt{a} \right\}, \quad z_\alpha = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right); \quad (6)$$

$$\text{disputed crash rate:} \quad A_\alpha(a) = \left\{ l_\alpha(a) \leq N_a \leq u_\alpha(a) \right\}, \quad (7)$$

with $\bar{X}_a$ the average growth observed, $\hat{\mu}$ the held drift, N_a the disaster count, and $\hat{\lambda}$ the held crash rate. The crash-rate endpoints have no elementary closed form— $A_\alpha(a)$ is the integer *Sterne region* (Sterne, 1954; Crow and Gardner, 1959), the smallest probability-ordered interval of held-model mass ($N_a \sim \text{Poisson}(\hat{\lambda}a)$) at least $1 - \alpha$, with explicit leading shape and step construction in Section 2.4 (Lemma D.1).

The interval form of these regions is not a modeling choice; it is the general form of an exact test of the held model: in either canonical channel the interval-acceptance tests form an essentially complete class, every other region weakly dominated and every admissible test almost everywhere an interval test (Lemma A.2, Appendix A; Birnbaum, 1955; Matthes and Truax, 1967). The form is forced, and the freedom that remains is where the endpoints sit. Exactness alone does not spend that freedom—it fixes the mass, and a further criterion places the endpoints: the classical unbiased selection adds a first-moment side condition (Lehmann and Romano, 2005, §4.2), while the probability-ordered selection (4)—the primitive’s—takes the highest-density region—Sterne’s—which the discrete channel forces to be conservative, of held-model mass at least $1 - \alpha$, since exact size on a lattice is available only by randomizing at the two endpoints. Either way one scalar remains, and the band’s shape in age—the $a^{-1/2}$ contraction of (6) (a $\sqrt{a}$ widening of the cumulative record’s band), the integer staircase of (7)—is inherited from the held model’s own fluctuation law, not chosen. The lemma characterizes each age’s acceptance set; running that per-age test at every age is the per-test rule itself. The primitive thus pins each *fixed-age* region; the dynamic objects—the first-exit time (5), its probation and right-continuity conventions, the hazard and its atoms—are the disciplined continuous-time embedding of that per-age test, the classical *repeated significance test* run continually (Armitage et al., 1969b; Siegmund, 1985), a convention adopted here rather than a limit theorem. The form is thus general; the dynamics are not: what a record can do—drift to its edge or jump across it—is channel content, and the results below are sharp because the two records are canonical.

Among size- α regions the primitive then pins the selection. Assembling exactness, the interval class, and the fine-resolution limit of Ortoleva’s cell rule yields the density-cutoff region—calibrated to the mass floor, the unique minimal region absent a flat

plateau, an interval by unimodality, the probability-ordered Sterne region in the counting channel; unimodality is the one assembly step, placing the cutoff region *inside* the complete class rather than merely dominated by a member of it. Proposition 3 (Appendix A) states the assembly with these regularity conditions. The selection is the primitive's, not exactness's: alternative exact-size selections—equal-tailed, unbiased—exist and are not generated by it.

(iii) What she does next. On abandoning her model she re-ranks the menu in light of the track record and adopts the new front-runner:

$$\rho'(\theta) \propto L_\theta(X_{(\sigma, \sigma+\tau]}) \rho(\theta), \quad \hat{\theta}' = \arg \max_{\theta} \rho'(\theta), \quad (8)$$

where $L_\theta(X_{(\sigma, \sigma+\tau]})$ is the likelihood under model θ of the just-ended regime's record, the segment from adoption σ across the holding time τ ; we abbreviate it X_σ where the regime is clear. The new model is judged only on data from its adoption onward—the *test record* restarts—while the *ranking* ρ carries forward and moves only at surprises. Re-ranking on each regime's record in turn equals re-ranking on the entire history at once (Appendix A).

Definition 1 (Surprise-governed belief revision). *Given a menu Θ with pairwise locally equivalent laws, an initial ranking ρ_0 , a tolerance $\alpha \in (0, 1)$, a recording resolution δ , and a fixed probation interval $a_0 \geq 0$, the investor holds $\hat{\theta}_t = \arg \max \rho_t$, initialized at $\arg \max \rho_0$; she monitors the held model through the summary T_{a_t} of its track record and abandons it at the first exit (5) from the ordinary range (4); and at each revision she re-ranks and re-adopts via (8), restarting the track record. The resulting process $(\hat{\theta}_t, T_{a_t}, a_t, \rho_t)$ —held model, monitored record, age, and ranking—is the surprise-governed belief process.*

Two features matter. The investor carries two memories: the track record she tests against restarts at each revision—a new model is judged on its own tenure—while the ranking ρ remembers everything, since learning which model is right is the work of a lifetime. The non-Bayesian feature is thus not a reset second-order prior (ρ accumulates full-history likelihood by (8)) but that the held model is the maximum-a-posteriori selection of ρ , not a posterior mean, revised only at exits of (5). Every clause of Definition 1 is Ortoleva’s one-shot rule carried into continuous time or forced by a degeneracy the alternative would create (Appendix A)—except one, which reflects a genuine choice.

That choice is how the tolerance α is interpreted once the model is tested again and again: as a *per-test* tolerance—the same size- α test at every instant, consulting only the held model—the literal continuous-time reading of Ortoleva’s rule, under which a model is abandoned when *its own* record looks too unlikely, not when a rival fits better. On that interpretation, the rule applies one *fixed* evidentiary standard that never rises with tenure, and the recurrence of Section 2.4 is its *signature*, not a defect of myopia.²

2.2 The record is a state variable

Once revision is governed by a test on the since-adoption record, the state a forecaster carries is her belief together with the record under which it has survived: the revision law—and the hazard derived from it in §2.3—is a coordinate of that enlarged

²The alternative reads α as a *lifetime* tolerance, bounding the probability of *ever* wrongly abandoning a correct model. Weinstein (2017) shows this bound is attainable by an α -spending rule and keeps the agent approximately dynamically consistent—the arbitrage against her is of order the lifetime rejection probability. But attaining it with power against the menu draws the test to the rival models’ likelihoods (Lemma A.8; by the admissibility characterization cited in its proof, any time-uniform guarantee rests on a nonnegative supermartingale), the model-switching rule of the paradigm-shift learner (Cho and Kasa, 2015; Ba, 2026), and forfeits surprise-governance. The per-test reading instead holds the size fixed at every instant, so the lifetime rejection probability is one: this is the regime the α -spending rule avoids, where that arbitrage bound is vacuous (Appendix A.7).

state, not recoverable from any *fixed-prior* Bayesian posterior on the public data. Its testable face is concrete: two forecasters reporting the same forecast today can revise differently tomorrow, their forecasts having survived different records.

Theorem 1 (The surviving record is a state variable). *Consider any rule of Definition 1 with a menu satisfying Assumption 1.*

- (i) *Insufficiency. The conditional law of the revision time (5) is not a function of $\hat{\theta}_t$ alone, nor measurable with respect to $\sigma(\beta_t)$ for any posterior β_t formed by updating a fixed prior on the public filtration $\mathcal{F}_t$.*
- (ii) *Sufficiency and irreducibility. It is a measurable function of $(\hat{\theta}, T_a, a)$, finite-dimensional by Assumption 1, and—when the band is nondegenerate in age and the monitored statistic discriminates the menu, as in both canonical channels—of no two of $\hat{\theta}, T_a, a$.*
- (iii) *Closure. Whenever the since-adoption statistic is autonomous—its law of motion determined by (T_a, a) alone, as in both canonical channels, whose stationary independent increments make the post-adoption record a fresh copy of the environment— $(\hat{\theta}, T_a, a, \rho)$ is a Markov process—piecewise-deterministic (Davis, 1984) in the crash channel, a boundary-reset diffusion in the trend channel—whose revisions are the successive first exits of T_a from (4), each post-revision ranking ρ' a function of (ρ, T_τ, τ) by the stopped sufficiency of Assumption 1 (cf. Küchler and Sørensen, 1997).*

Each part has a settled lineage; the rule of Definition 1 is what joins them. Carrying the age since a reset to recover a Markov state is the method of supplementary variables (Cox, 1955) (parts i, iii); the collapse of the record to a finite summary (T_a, a) is the recursive sufficiency of an exponential family (Küchler and Sørensen, 1989) (part ii); the joined state—revision its first exit—is, in the crash channel, a

piecewise-deterministic process (Davis, 1984), deterministic between disasters, and in the trend channel a boundary-reset diffusion, both closed under the menu (Küchler and Sørensen, 1998) (part iii). What is new is their composition: the age is the time since the last *model abandonment*, the sufficient statistic is the *test record* the held model is judged on, and the first exit is its *rejection*. The one place this leaves classical ground is part (i)—the belief is not a sufficient statistic for its own revision—inverting the Bayesian benchmark, where the posterior is its own state (Epstein and Le Breton, 1993).

Part (i) is where the two forecasters of the introduction part: sharing every public-history posterior, they still face different hazards, because the hazard turns on the private adoption date σ (equivalently the age $a = t - \sigma$) that no public-filtration posterior carries. The enlargement to $(\hat{\theta}, T_a, a)$ is a property of the record-resetting rule itself, not of the Sterne-band implementation below.

Part (iii) licenses the dynamics that follow: because the enlarged state is a bona fide Markov process, its evolution between and at revisions, its long-run behavior, and its cross-sectional law are all well-posed objects rather than artifacts of an ill-specified state.

The economic content is that a belief carries a hidden coordinate. Two investors agreeing on the held model need not agree on how fragile that agreement is: the one whose record sits at the edge of its band revises on the next surprise, the one in the interior does not. Consensus can thus mask fragility—a population agreeing on a forecast may be uniformly robust or one surprise from fracture, and the level alone cannot tell them apart. The prediction is cross-sectional and testable on forecaster panels (the Survey of Professional Forecasters, the Michigan survey): conditional on the same current forecast, the probability of revising should depend on regime age and run length—how long the forecast has been held and the cumulative surprise it

has absorbed—not on the level. By Lemma G.1 this is an implication for the *observed* hazard conditional on the full inference state (R_t, a_t, ρ_t) : the self-re-adoption thinning that makes reported revisions sparse is itself stakes-free, but it also lets a reported spell differ from the internal regime age, so the panel test conditions on reported duration and cumulative surprise as *proxies*—exact when self-re-adoptions are absent, a structural restriction rather than a regression prediction otherwise.

What the theorem excludes, exactly: the held model alone, the reported forecast, the pair of model and spell age without the record’s position, and every posterior formed from a *fixed* prior on the public filtration. What it does not exclude is the enlarged state itself; and the carried ranking ρ_t —likelihood-based but frozen between revisions—enters the *re-selection*, not the pre-exit hazard, so consulting it is not a hidden-state relabeling of the belief.

What the theorem fixes is what the state *is*; the economics is in what it *does*. Ortoleva’s rule is a single test—reject, re-select, and stop—with no dynamics to speak of. Set it running in continuous time and the record it monitors acquires a life the belief does not share. Between surprises the state *drifts*: the record ages and the band it is tested against moves, so a model’s proximity to revision changes while the reported belief holds still. At a revision it *resets*—record and age restart, and the newly adopted model begins with a clean, robust record whatever it replaced. Left to run, the two motions never resolve: the model’s *location* consolidates, on the truth or the nearest rival, while its record keeps crossing out of band, so revision never stops even after the belief has stopped moving. And a population of such records, watching one path, moves as a single *flow*—the cross-section carried forward as the image of the fixed distribution of tolerances.

None of these is a motion of the belief; each is a motion of the coordinate the belief omits. That is the substance of the enlargement, and it is what a belief-based account

cannot supply: where persistence lives in the posterior, two agents who share it share their futures, whereas here the moving object is one the posterior does not carry, so a common belief is consistent with divergent fragilities and divergent paths to revision. The rest of this section traces that one moving coordinate—the tempo of its drift to reset (§2.3), the permanence of the cycling (§2.4), and the heavy tail of that tempo (§2.5)—before Section 3 follows it across the market.

2.3 The revision hazard

The enlarged state of Theorem 1 governs a rate: the rate at which the investor abandons her model, the endogenous belief hazard the title names. Its absolutely continuous part is

$$h(a) = \lim_{da \downarrow 0} \frac{1}{da} \mathbb{P}(\tau \in (a, a + da] \mid \tau > a), \quad (9)$$

the age hazard of the holding time τ of (5). In the crash channel the band endpoints (7) step at discrete ages, so revision also carries point masses and the full object is a *measure*—the continuous hazard (9) together with atoms at the scheduled band-step dates (Appendix C).

The hazard is not a primitive: it is taken from the first-exit law of the rule (4)–(5), not posited. One distinction the measure hides drives the enlargement. The rate $h(a)$ is that of the *marginal* holding-time law—the law of τ averaged over records—so it is a function of the reported coordinates $(\hat{\theta}, \alpha, a)$ —per tolerance fibre—and closes there. The *conditional* exit propensity given the full state—how close the record T_a sits to the band edge—is a different object: two age- a holders of the same $(\hat{\theta}, \alpha)$ share $h(\hat{\theta}, \alpha, a)$ yet, sitting at different T_a , stand at different distances from revision. That gap, between the marginal hazard on $(\hat{\theta}, \alpha, a)$ and the conditional fragility on

the hidden T_a , is the insufficiency of Theorem 1(i) expressed on the hazard itself. The split takes a different form in each channel—a state-dependent jump rate at the inaccessible crash exits, a predictable boundary flux in the trend channel, a mixed compensator—and Appendix C records it together with the measure above.

This history-dependent hazard is the belief-side counterpart of the adjustment hazard that organizes the state-dependent pricing literature (Caballero and Engel, 1999; Alvarez et al., 2022)—endogenous like theirs, but governed by surprise rather than the cost of acting. The constant Calvo (Calvo, 1983) and sticky-information (Mankiw and Reis, 2002) hazard is not attainable by the canonical rules—their correct-model hazard decays like $\theta(\alpha)/a$ (Corollary 1)—and enters only as the reduced-form baseline λ_0 of the hybrid nesting (Definition 2), the record-free member of the class. Since state- and time-dependent adjustment rules nearly coincide to first order (Auclert et al., 2024), the departure we derive rather than assume is not first-order but in the tail and its timing—the heavy-tailed holding time of §2.5, the recurrence of §2.4, and the crash channel’s directional asymmetry—none of which the first-order equivalence reaches.

Within this class the rule is not one flexible member among many; it is the member that adds no primitive beyond a scalar. A cost-based hazard is microfounded in optimization, and its shape is priced off a cost distribution that is itself a free object, calibrated to fit. The testing hazard is microfounded in inference, and its shape is inherited from an object the model already carries—the null’s own fluctuation law—so, given the recording environment, the entire hazard family is spanned by the single scalar α . The primitive is axiomatized (Ortoleva, 2012); the per-test size is its literal continuous-time reading, the lifetime reading and its approximate dynamic consistency (Weinstein, 2017) being the alternative of Appendix A.7; and the shape of the hazard is a prediction rather than a calibration target. Stakes-invariance is then

the separating test between the two microfoundations of the one class—its empirical use an identification problem, the restriction conditioning on the hidden inference state (Appendix F).

Every non-representability statement below is relative to a named class—a fixed-prior Bayesian’s held belief, the conditional expectation given $(\mathcal{F}_t)$ of a single prior over the menu—and is established over a positive time interval, never by conditioning on an instantaneous event.

2.4 Location consolidates, activity does not

Two forces act on a held model over a long career. The ranking ρ accumulates evidence and so its argmax settles; the test, applied afresh at every instant, keeps finding the held model surprising. The first force is consolidation, the second recurrence, and they do not cancel: the model a believer holds stops changing in *identity* while never ceasing to change in *tenure*.

Proposition 1 (Location consolidates, activity does not). *Run the rule of Definition 1 at a fixed tolerance $\alpha \in (0, 1)$ in either canonical channel—the disputed-trend record (a Brownian running mean) or the disputed-crash record (a Poisson count)—and suppose the menu has a unique closest model in relative-entropy rate, $\theta^\circ = \arg \min_\theta \kappa(\theta)$ with $\kappa(\theta) := \lim_{t \rightarrow \infty} t^{-1} \text{KL}(\mathbb{P}_t^* \| \mathbb{P}_t^\theta)$ —the finite object here, the full-path divergence being infinite—with $\rho_0(\theta^\circ) > 0$.*

- (a) Location. *Under the chained update (Appendix A.4) there is an a.s. finite calendar time T° such that $\sigma_k \geq T^\circ \Rightarrow \arg \max_\theta \rho_{\sigma_k}(\theta) = \theta^\circ$ almost surely.*
- (b) Activity. *For any held model the holding time τ of (5) is $\mathbb{P}^*$ -almost surely finite, and, re-adopting the model after each exit, the agent abandons it infinitely often, $\mathbb{P}^*$ -almost surely. When the menu contains the truth, every regime begun after*

T° starts with θ° and is abandoned again.

Both halves are established facts; the rule of Definition 1 is what holds them together. Part (a), proved by the strong law for the log-likelihood ratio, makes no reference to whether or when a revision occurs—that is part (b). It places the held-model *location* where the misspecified-learning literature places the *limit* of belief: the Kullback–Leibler projection of the truth (Berk, 1966; Esponda and Pouzo, 2016; Fudenberg et al., 2021; Frick et al., 2023). When the truth is on the menu, θ° is the truth and the ranking *merges* with it—the condition $\rho_0(\theta^\circ) > 0$ is the grain of truth (Kalai and Lehrer, 1993) under which opinions merge (Blackwell and Dubins, 1962), which is why positive weight on the closest model, not full support, suffices for part (a). Any residual gap under a misspecified menu is the menu’s, not a dynamic error of the testing, and at a tie the front-runner consolidates to the closest set and cycles within it.

The economic content is that a correct model is right on average yet locally surprising: tested at a fixed size and monitored without limit, its own ordinary fluctuations eventually carry the record out of band. This is “sampling to a foregone conclusion”—a fixed-level test on accumulating data rejects a true null with certainty, by the boundary-crossing of the law of the iterated logarithm (Anscombe, 1954; Armitage et al., 1969a; Robbins, 1970; Robbins and Siegmund, 1970)—here the engine of the dynamics rather than a defect. In its modern form this is the object that safe, anytime-valid inference is built to correct: holding the test size fixed at every instant is exactly what test-martingales and e-processes avoid (Ramdas et al., 2023; Grünwald et al., 2024), and it is what sends the lifetime rejection probability to one. Our agent forgoes that correction: the canonical rules attaining a lifetime bound mix over the rival models (Lemma A.8), the one thing surprise-governance gives up—so the

recurrence is the signature of that choice. It rests on the fluctuation law of each record, not on the Markov closure of Theorem 1 alone; we do not assert recurrence for an arbitrary time-homogeneous data-generating process. So a correct-on-average belief is revised forever—correctness buys no permanence—and a correct-on-average population keeps revising, its members agreeing on the model yet never on how close each stands to abandoning it (Section 3). We claim almost-sure finiteness and infinite recurrence, not a finite expected holding time, which in the trend channel can fail (§2.5).

The argument runs through a single object. In the trend channel the standardized statistic is, in logarithmic age, a stationary Ornstein–Uhlenbeck process (the Lamperti transform of the running mean); its exit set carries positive invariant mass, so an ergodic diffusion enters it in finite time and returns forever—finite τ and recurrence at once. The centered Poisson count plays the same role in the crash channel, its $\sqrt{a \log \log a}$ fluctuations outgrowing the band’s $O(\sqrt{a})$ half-width. Independent restarts (Appendix A.5) whose holding times cannot accumulate (Lemma D.2) then turn each a.s.-finite regime into infinitely many revisions. Full proofs, including the bridge across the regime in force at consolidation, are in Appendix D.

2.5 The holding-time law

Recurrence says a model is abandoned; the holding-time law says when. In the crash channel the survival has an exact finite matrix-product form—an age-inhomogeneous phase-type construction, the band dimension changing across step ages (Lemma D.3, Appendix D.4): it is the ex-ante single-spell law, distinct from the pathwise push-forward that moves the realized cross-section of Section 3. What the population dynamics inherit is its *tail*, and that tail is heavy in both channels—heavy enough

that the mean quiet spell can be infinite, the continuous-monitoring descendant of Blackwell and Freedman (1964)'s observation that the two-sided $c\sqrt{n}$ exit of a coin-tossing record has finite mean only for $c < 1$. The recurrence itself is the continuous-monitoring form of the classical fact that repeated significance testing at fixed size rejects a true hypothesis with probability one (Armitage et al., 1969b; Siegmund, 1985).

Corollary 1 (Holding-time law). *In the setting of Proposition 1, where the holding time τ is a.s. finite: (a) holding the correct model, one exponent governs the tail of τ in both channels—the principal Dirichlet eigenvalue $\theta(\alpha) > 0$ of the Ornstein–Uhlenbeck generator $\frac{1}{2}\partial_{vv} - \frac{1}{2}v\partial_v$ on $(-z_\alpha, z_\alpha)$: in the trend channel the tail is an exact power law,*

$$\mathbb{P}(\tau > t) \sim C t^{-\theta(\alpha)} \quad (t \rightarrow \infty), \quad (10)$$

and in the crash channel

$$\mathbb{P}(\tau > t) = t^{-\theta(\alpha)(1+o(1))} \quad (t \rightarrow \infty), \quad (11)$$

so the mean quiet spell is finite when $\theta(\alpha) > 1$ and infinite for sufficiently tolerant investors (wide bands, $\theta(\alpha) < 1$); (b) holding an incorrect model, τ is finite, with, conditional on no earlier false alarm, detection scale

$$a_{trend}^* \approx \frac{z_\alpha^2 \sigma^2}{(\mu^* - \hat{\mu})^2}, \quad a_{crash}^* \approx \frac{z_\alpha^2 \hat{\lambda}}{(\lambda^* - \hat{\lambda})^2}, \quad (12)$$

the scale growing as the inverse square of the misfit, so larger misfits are detected sooner—in the local-misspecification regime an order-one estimate of the scale, $\log \tau / \log a^ \rightarrow 1$ conditional on no earlier false alarm, not a sharp limit—so that for two held models equidistant in Kullback–Leibler rate from the truth the conditional detection scales*

agree to leading order, $a^* \sim z_\alpha^2/(2K)$ in both channels, the narrower band detecting faster only at the next order.

Which of these conclusions depend on the Sterne selection? Only the calendar. Any exact-size nested family of intervals with the same asymptotic standardized endpoints $\mp z_\alpha$ —and in the symmetric limits of both channels the natural selections, highest-density, equal-tailed, central, share them—has the same tail exponent, the same finite/infinite-mean threshold, and the same recurrence: the trend problem by a domain sandwich and the continuity of λ_0 , the crash tail because the exit lemma’s hypotheses consume only the asymptotes. What moves with the selection is the schedule—the step dates, the predictable atoms, the finite-age band geometry (Lemma D.7, Appendix D.5).

One killed principal eigenvalue $\theta(\alpha)$ governs both the recurrence of Proposition 1(b) and the heavy tail—the trend statistic is the Ornstein–Uhlenbeck process exactly, the standardized crash count converges to it—with the crash channel carrying its survival through the matrix-product law of Lemma D.3. The two parts are the rule’s *error profile*: a correct model held long, a wrong one caught fast, and widening the band lengthens both—tolerance trades false alarms against detection lag.

In the crash channel the *direction* of a model’s error fixes *how* it dies—conditional on no earlier false alarm, the detection-scale regime of Corollary 1(b) and Proposition 4. A complacent model, understating the disaster rate, then dies the moment one disaster too many arrives—an event-triggered up-jump in perceived risk, with no warning. A fearful model, overstating it, dies only once a stretch of calm has run conspicuously long, at a date one could have circled in advance—a scheduled downgrade. Conditional on detection, revisions up are surprises and revisions down are scheduled; an early fluctuation can still kill either model at either boundary. The trend channel, a

diffusion crossing its band in both directions, has no such asymmetry; and throughout, the world’s disaster rate never moves—what spikes and bleeds is the *perceived* hazard. Proposition 4 (Appendix D.7) gives the directional map and its scope.

3 The cross-section is a pushforward

A single believer has a survival law; a market has a population of them. Because investors watch the same data and differ only in their tolerance for surprise, the cross-section of beliefs is the *exact* image of the tolerance law under the common flow—no law of large numbers, no equilibrium average to solve for—so the aggregate dynamics carry no *behavioral* parameter beyond the test size α ; the heterogeneity they carry enters through the tolerance law G , a primitive that quantitative work must specify or estimate, not a parameter tuned to the patterns. This section states that pushforward and derives its immediate dynamics from it; the cross-sectional and asset-pricing patterns it implies—a selected consensus, endogenous disagreement, a disaster-fear saw-tooth, experience effects—are developed in the working-paper version.

Index an investor by a type $\xi \in \Xi$ —a tolerance $\alpha(\xi)$, and a fixed menu and prior—drawn from a cross-sectional law $G \in \mathcal{P}(\Xi)$. All investors observe the same path of the economy’s process X and run the rule of Definition 1 on it; types differ only in the band width $z_{\alpha(\xi)}$, so they exit and re-select at different times along the one path.

Assumption 2 (Common path). *Investors observe the same public record and differ only in tolerance, holding a fixed menu and prior; the cross-section carries no idiosyncratic agent-level noise. The tolerance law G has support bounded away from the degenerate band— $\alpha(\xi) \leq \bar{\alpha} < 1$ for G -a.e. ξ —so every type’s standardized critical value is bounded below, $z_{\alpha(\xi)} \geq z_{\bar{\alpha}} > 0$; the raw band width still varies with age, contracting like $a^{-1/2}$ in the trend statistic, with single-integer Sterne regions possible*

at early ages. In the trend channel the fixed probation onset of Definition 1—not the support bound on α —gives each type the deterministic floor $\tau_k \geq a_0 > 0$; in the crash channel holding times are a.s. positive but not deterministically bounded below ($a_0 = 0$, since a disaster can carry an upper exit arbitrarily close to adoption), and local finiteness of the aggregate flux comes channel-specifically: finitely many disasters on compacts, and a uniform bound on the G -average number of scheduled floor-step exits per compact—the flux measure locally finite, not the population-level step dates finitely many (proof of Corollary 2).

The model’s one structural commitment for aggregation is that all cross-sectional heterogeneity sits in the test threshold, not in information or private beliefs (the overlapping-generations extension adds birth date as the only other axis). This is what makes the pushforward exact—the cross-section is the image of a primitive, not a calibrated object, so which patterns appear follows from G and the environment, not from a tuned parameter; G itself is the object a quantitative application must specify or estimate. We maintain Assumption 2 throughout; two further *environment-specific* assumptions—local misspecification (Assumption 3) and demographic turnover—are invoked only where needed. The results below divide by what they condition on. Proposition 2 is an identity in G : given the common path it holds for every tolerance distribution and at every date—a statement about the *map* from primitive to cross-section, valid along any transition path, presuming no stationarity. The statements that follow are conditional exactly as their hypotheses say: turnover and the weighting license the terms of Corollary 2; atomless G governs which parts of the flux are continuous; the local-misspecification regime (Assumption 3) scopes every re-selection-direction claim; and quantitative patterns are G -conditional—they follow once G is specified and change with it. Statements about stationary cross-sections are

separate results with their own existence hypotheses, not consequences of the push-forward identity. The aggregate-jump claims are additionally conditional on (17) together with a positive-measure model-changing crossing set, verified per application. Two revision counters run through the paper: the *internal* trigger—every band exit, self-re-adoptions included—whose recurrence Proposition 1 establishes; and the *observed* model-changing count, its (U)-thinning (Lemma G.1). Past consolidation with the truth on the menu, exits recur while observed changes can vanish; every aggregate statement that turns on observable revision is stated in the thinned counter. Write $s_t(\xi) = (\hat{\theta}_t, \alpha(\xi), a_t) \in \mathcal{S}$ for the investor’s *held-model state* at t —the held model, the type’s tolerance, and the age of its test record, $\mathcal{S} = \Theta \times (0, 1) \times \mathbb{R}_+$; the tolerance is a coordinate of the state because it sets the band the record is tested against, and it is constant along each path ($\dot{\alpha} = 0$), so it adds no transport term and the flow acts fibrewise in α . Write $\bar{s}_t(\xi) = (\hat{\theta}_t, \alpha(\xi), a_t, R_t, \rho_t) \in \bar{\mathcal{S}}$ for the *full* belief state, which adjoins the since-adoption record summary $R_t = T_{a_t}$ that drives the boundary and the carried-forward ranking ρ_t , with $\bar{\mathcal{S}} = \mathcal{S} \times \mathcal{R} \times \mathcal{P}(\Theta)$. Let $\bar{\Psi}_t : \Xi \rightarrow \bar{\mathcal{S}}$ be the flow the common path induces on full states and $\Psi_t = \pi \circ \bar{\Psi}_t$ its held-model projection, $\pi : \bar{\mathcal{S}} \rightarrow \mathcal{S}$. The cross-section of beliefs we report is the held-model marginal

$$m_t := \Psi_t \# G = \pi \# \bar{m}_t \in \mathcal{P}(\mathcal{S}), \quad \langle \varphi, m_t \rangle = \int_{\mathcal{S}} \varphi dm_t = \int_{\Xi} \varphi(\Psi_t(\xi)) G(d\xi), \quad (13)$$

the projection onto $(\hat{\theta}, \alpha, a)$ of the full-state cross-section $\bar{m}_t := \bar{\Psi}_t \# G$; where an aggregate uses only $(\hat{\theta}, a)$ it takes the further marginal, but the cross-section itself carries the tolerance coordinate throughout. The aggregates we form—fear, dispersion, the consensus—are test functions $\varphi(\hat{\theta}, a)$ of these coordinates, so m_t carries them. In particular the *belief functional* $w(\hat{\theta})$ —the value the held model reports, the perceived

intensity $\hat{\lambda}$ in the crash channel and the perceived drift $\hat{\mu}$ in the trend—is one such test function, so the consensus belief $\langle w, m_t \rangle$ and the cross-sectional disagreement $\text{Var}_{m_t}(w)$ used below are carried by m_t . But the boundary-crossing and re-selection that *move* the cross-section depend on the record and ranking (R_t, ρ_t) , which are not coordinates of $\mathcal{S}$. The forward equation of §3.1 is for that reason the *pathwise* time-derivative of the deterministic pushforward $\Psi_t \# G$ (Proposition 2), not a closed generator equation on $\mathcal{S}$: the full state on $\bar{\mathcal{S}}$ is Markov—piecewise-deterministic in the crash channel, a diffusion with boundary reset in the trend—but along the common path (R_t, ρ_t) are deterministic functions of ξ and t , so the projected dynamics follow from the pushforward.

Proposition 2 (Pushforward cross-section). *For every bounded measurable φ , (13) holds pathwise: under Assumption 2, m_t is the exact image of the tolerance law G under the common flow Ψ_t , with no continuum law of large numbers invoked. The realized non-revision fraction at t —the G -mass of types whose held model is not revising in the instant at t —is a pathwise functional of X , an instantaneous quantity, distinct from the survival of a holding spell from adoption to age a . The ex-ante population survivor fraction to age a is the G -mixture $\int_{\Xi} S_{\xi}(a) G(d\xi)$ of the single-type laws (C.2), not the single-type $S(a)$; the realized and ex-ante objects agree only in expectation over paths.*

The force of Proposition 2 is that heterogeneity, not noise averaging, generates the cross-section: each type is a deterministic function of the same history, so the belief distribution is a change of variables away from the tolerance distribution, and every later statement can be made pathwise without solving for an equilibrium average.

3.1 What moves an aggregate

Between revisions a belief state only ages; at a revision it jumps. Lifting this to the population along the common path—differentiating the deterministic pushforward $\Psi_t \# G$ (Proposition 2)—gives the forward equation for any index—three channels on the head count, (14), and a fourth, selection, on the wealth weighting, (15).

Corollary 2 (Four channels). *Let $\varphi \in C_b^1(\mathcal{S})$. Along the common path, for the cross-section m_t with overlapping-generations turnover—death at rate p independent of the record, birth into the type’s newborn state $s_\circ(\xi) = (\arg \max_\theta \rho_0(\theta), \alpha(\xi), 0)$, each agent reborn with her own tolerance, the newborn cross-section $m_\circ := s_\circ \# G$, the closed population being the case $p = 0$ —the head-count average obeys*

$$d\langle \varphi, m_t \rangle = \underbrace{\langle \partial_a \varphi, m_t \rangle dt}_{\text{transport}} + \underbrace{\int (\varphi \circ J - \varphi) d\Phi_t}_{\text{revision flux}} + \underbrace{p(\langle \varphi, m_\circ \rangle - \langle \varphi, m_t \rangle) dt}_{\text{turnover}} ; \quad (14)$$

and, in the closed population ($p = 0$) under the wealth weighting of (E.1), the wealth-weighted average obeys, between common disasters—the weights’ motion is continuous there; at a common disaster wealth shares and beliefs jump together, the joint atom being the (17) object, outside this display—

$$d\langle \varphi, m_t^w \rangle = \langle \partial_a \varphi, m_t^w \rangle dt + \int (\varphi \circ J - \varphi) d\Phi_t^w + \underbrace{d\Sigma_t(\varphi)}_{\text{selection}}, \quad (15)$$

the same integrals under the tilt, Φ_t^w the flux weighted by the crossers’ wealth shares, plus the selection term of (E.1), where ∂_a is the ageing drift; Φ_t the crossing flux—a measure over the pre-jump full states (equivalently, over crossing types, so that $\varphi \circ J$, which consults the stopped record and carried ranking, is well defined) of held models reaching their band edge (4) along the common path, recording where each crosses so

that $\int(\varphi \circ J - \varphi) d\Phi_t$ is well defined for the state-dependent increment—and J the re-selection-and-restart map (8) of the crossing mass; p is the turnover rate, $m_\circ$ the newborn cross-section, and Σ_t the selection term of (E.1).

The four channels—three on the head count, selection on the wealth weighting between common disasters—are the exhaustive list of why a consensus belief moves: records age (transport), investors revise (flux), the population is replaced (turnover), and—if beliefs are wealth-weighted for pricing—the weights drift with relative performance (selection). The first two carry the enlarged state’s record coordinates—transport ages the record, flux is its crossing—while turnover and selection would act on any population; and it is the flux, whose size turns on where each crosser’s record sits, that fails to close on the reported marginal m_t , the population image of the single-agent insufficiency (Theorem 1). The split inside the flux is the disaster asymmetry of §2.5 at the population level: scheduled downgrades at *predictable* times, disaster-driven upgrades at *inaccessible* ones.

The flux Φ_t is in general not Lebesgue-absolutely-continuous; at a common disaster it carries an *inaccessible atom* $\Delta\Phi_u = b_u$, the crossing mass, while dispersed trend crossings and disaster-synchronized floor-steps supply its continuous and predictable parts (Appendix E).

Corollary 3 (Boundary-mass sufficient statistic). *At a common-surprise date u , the jump in any linear index is the integral of the test-function increment over the crossing mass alone,*

$$\Delta\langle\varphi, m_u\rangle = \int(\varphi \circ J - \varphi) db_u, \quad (16)$$

so the aggregate response is a functional of the boundary mass b_u as a measure—one that records each crosser’s re-selection image J —rather than of its total occupation alone: the effective driver is the sub-mass whose re-selection moves the test function

$(\varphi \circ J \neq \varphi)$: for a belief-only index $\varphi = w(\hat{\theta})$ this is the model-changing sub-mass, self-re-adopters contributing zero, while for an age-dependent φ the record restart ($a \rightarrow 0$) makes them contribute. For a nonlinear aggregate the response is the full increment, not Γ' times (16).

Equation (16) is the belief-side counterpart of the sufficient statistic governing aggregate adjustment in the generalized-hazard literature (Caballero and Engel, 1999; Alvarez et al., 2022): there the response to a shock is carried by the mass at the adjustment threshold, here by the mass at the test boundary. Only investors whose record is *at* the band edge when the surprise lands revise; the interior does not move. The size of a fear wave is set at the boundary—by the crossing mass and where its re-selection sends it, not by scalar occupation. A complacent market is not for that reason safe: when the consensus has settled below the true disaster rate, the count driven by the true rate overtakes the too-low ceilings, leaving a mass poised at the boundary—a latent fragility the *level* does not reveal, since two consensuses at the same level can carry different masses poised to revise—so an ordinary disaster moves the aggregate sharply where, against a consensus with its mass interior, it would barely register.

We name the condition this isolates—that a common surprise carry positive boundary mass, an *environment-specific* condition on the population’s record configuration, not a behavioral parameter—and invoke it in the applications below. Writing ∂ for the band-edge set (the upper edge $\{T_a = u_\alpha(a)\}$ in the crash channel) and b_u for the crossing measure at u ,

$$(B+) \quad b_u(\partial) > 0 \quad \text{at the surprise date } u \quad (17)$$

requires positive mass *crossing* at u , not merely occupying a boundary set before-

hand. Absent (B+) a common disaster moves the aggregate only through the smooth channels; with it (and provided a positive-measure set of the crossing mass re-selects to a different rate) the crash channel delivers a jump the trend channel cannot—its crossings dispersed across an atomless G with no synchronizing common date; with atoms in G , or trend crossings synchronized by a shared reset age, the contrast is between inaccessible and predictable jumps rather than jump versus none.

These are the aggregate’s moving parts; their consequences—for the consensus, for disagreement, and for the disaster, overlapping-generations, and transition-path economies—are worked out in the working-paper version.

4 Conclusion

One rule does the work of this paper. An agent keeps a single model, tests it against its own record at a fixed size, and abandons it the first time the record looks too unlikely. From this one primitive—no bias, no noise, no menu cost, one free *behavioral* parameter, the test size—follows an exact, history-dependent rate of model abandonment, the endogenous belief hazard, governed by a state the belief does not contain: the held model *enlarged* by its surviving record, so two forecasters reporting the same belief can carry different fragilities. The rest is consequence, beginning with recurrence: even a correct model is abandoned again and again, its location consolidating to the truth—or the nearest menu model—while revision activity never stops.

Aggregated over a population, the primitive delivers the cross-section’s texture from one mechanism. The cross-section is the exact pushforward of the tolerance distribution; individual beliefs are pure-jump while the consensus, averaging staggered jumps across a continuum, is continuous away from the common surprises that synchronize model-changing crossings—micro-lumpy, macro-sluggish, two reaction findings

at two aggregations. Disagreement is cyclical, jumping when common surprises synchronize *model-changing* crossings and narrowing through the calm; with demographic turnover it is non-monotone in the turnover rate—a signature a behavioral-dispersion account has no reason to produce. And a date-0 surprise leaves a belief overhang decaying as $e^{-(p+\kappa_{\text{rev}})t}$ —half-life near fourteen years at conventional rates—so on a thirty-year sample a stationary process wears an *informational* secular trend, the mechanism supplying the timescale.

Taking the model to the economy runs on sign and timing. In the rare-disaster economy the consensus fear jumps up at each disaster carrying model-changing boundary mass and bleeds down only on scheduled dates, the true intensity and preferences held fixed; under any monotone kernel the jump-risk premium inherits this sawtooth—countercyclical, its spikes sustained by turnover’s replenished complacency. The stand-in kernel supplies the monotonicity and pins the level; the model’s own claims are the timing and the sign. These application results are derived in the Online Appendix.

We have been deliberate about scope. The belief dynamics are proved from the rule; the price-level claims are sign and timing, conditional on a monotone kernel, the level deferred to the companion paper. Aggregate phenomena that turn on boundary mass require positive crossing mass—condition (17)—with model-changing re-selection on a positive-measure subset of it—condition (U)—whose primitive characterization we leave open. Because the trigger is the record’s improbability, not the payoff, the hazard is stakes-invariant—the separating restriction against cost-based rivals (Cohen et al., 2018), its test an identification problem—the restriction conditions on the hidden inference state—outlined rather than solved (Appendix F).

The direction we find most promising is the endogenous case. Throughout, the record each investor tests is exogenous—the common path—while in a traded equilibrium

records are made by beliefs, weighted by wealth, and moved by the very disasters the agent tests for: the tested environment becomes, in part, the population's own output. The companion equilibrium paper closes that loop for the disaster economy—market clearing, the wealth-weighted belief multiplier, the no-arbitrage gains process. The same endogeneity runs naturally through policy and the real economy wherever either responds to the consensus belief and thereby shapes the record that revises it. Whether the primitive keeps delivering disciplined dynamics once its data are endogenous is the open question that matters most; that it can be posed without changing the rule is the sense in which the primitive is a foundation rather than a reduced form.

A Grounding of the continuous-time rule

This appendix supplies the results that Section 2 points to: each clause of Definition 1 is either the continuous-time transcription of Ortoleva's static rule or forced by a degeneracy the alternative would create. The lemmas appear in the order in which the body uses them. For the two worked channels we take, under a model θ , either the disputed-trend law

$$dX_t = \mu dt + \sigma dW_t, \quad \theta = \mu \in \{\mu_1, \dots, \mu_K\} \subset \mathbb{R}, \quad \sigma > 0 \text{ known}, \quad (\text{A.1})$$

with W a $\mathbb{P}^\theta$ -Brownian motion and summary $T_a = \bar{X}_a := X_a/a$; or the disputed-crash-rate law

$$N \text{ a Poisson process of rate } \lambda, \quad \theta = \lambda \in \{\lambda_1, \dots, \lambda_K\} \subset (0, \infty), \quad (\text{A.2})$$

with summary $T_a = N_a$. Write $\mathbb{P}^\theta$ for the law on $(\mathcal{F}_t)$ and $L_\theta(\cdot)$ for the likelihood of a record under θ .

A.1 Disputability of the menu

Lemma A.1 (Disputability). *For the data (A.1)–(A.2) observed continuously on $(\mathcal{F}_t)$: (i) distinct drifts $\mu \neq \mu'$ induce mutually equivalent laws on $\mathcal{F}_t$ for every $t > 0$, with density*

$$\left. \frac{d\mathbb{P}^{\mu'}}{d\mathbb{P}^\mu} \right|_{\mathcal{F}_t} = \exp\left(\frac{\mu' - \mu}{\sigma^2} (X_t - \mu t) - \frac{(\mu' - \mu)^2}{2\sigma^2} t\right); \quad (\text{A.3})$$

(ii) distinct rates $\lambda \neq \lambda'$ induce mutually equivalent laws on $\mathcal{F}_t$ for every $t > 0$, with density

$$\left. \frac{d\mathbb{P}^{\lambda'}}{d\mathbb{P}^\lambda} \right|_{\mathcal{F}_t} = \left(\frac{\lambda'}{\lambda}\right)^{N_t} e^{-(\lambda' - \lambda)t}; \quad (\text{A.4})$$

(iii) two laws differing only in the diffusion coefficient $\sigma \neq \sigma'$ are mutually singular on $\mathcal{F}_t$ for every $t > 0$. Hence the drift and the jump intensity are disputable on the menu, the diffusion coefficient is not.

Proof. (i)–(ii). The densities (A.3)–(A.4) are the Girsanov and Doléans–Dade exponentials of a drift and an intensity change. For the trend law the exponential of $\gamma W_t - \frac{1}{2}\gamma^2 t$, $\gamma = (\mu' - \mu)/\sigma$, is a true $\mathbb{P}^\mu$ -martingale—Novikov’s condition $\mathbb{E}^\mu \exp(\frac{1}{2}\gamma^2 t) < \infty$ holds—so $d\mathbb{P}^{\mu'} = Z_t d\mathbb{P}^\mu$ is a probability equivalent to $\mathbb{P}^\mu$ under which, by Girsanov, $W_t - \gamma t$ is a Brownian motion and X has drift μ' (Karatzas and Shreve, 1991, §3.5). For the crash law the exponential $(\lambda'/\lambda)^{N_t} e^{-(\lambda' - \lambda)t}$ has bounded jump integrand $\lambda'/\lambda - 1$, hence is a true martingale, and under $d\mathbb{P}^{\lambda'} = Z_t d\mathbb{P}^\lambda$ the compensated count $N_t - \lambda't$ is a local martingale, so N is a $\mathbb{P}^{\lambda'}$ -Poisson process of rate λ' —a genuine intensity change, not merely a match of one-dimensional marginals (Brémaud, 1981). Positivity $Z_t > 0$ gives equivalence in both directions. (iii). The

realized quadratic variation $[X]_t$ equals $\sigma^2 t$ $\mathbb{P}^\sigma$ -a.s. and $\sigma'^2 t$ $\mathbb{P}^{\sigma'}$ -a.s. (Karatzas and Shreve, 1991, §1.5), distinct for $t > 0$, so the two laws are mutually singular and the diffusion coefficient is identified pathwise, beyond dispute (Merton, 1980). For menus beyond (A.1)–(A.2) the same dichotomy—drift and intensity equivalent, diffusion coefficient singular—is the content of the jump-diffusion criteria of Cheridito et al. (2005), whose hypotheses (a bounded generator transformation, here the constant drift- or intensity-shift, and a well-posed martingale problem for the target) hold here; only the explicit densities (A.3)–(A.4) are used below. $\square$

The lemma fixes what a menu may dispute: the held model in (1) ranges over drifts or intensities, with the diffusion coefficient common and known, so the densities L_θ in (8) are well defined and the track record has a likelihood under each candidate.

Lemma A.2 (Band form). *Fix an age $a > 0$ and a held model $\hat{\theta}$, and consider testing $H_0 : \theta = \hat{\theta}$ against $H_1 : \theta \neq \hat{\theta}$ on the basis of T_a , in either canonical channel: $\bar{X}_a \sim \mathcal{N}(\theta, \sigma^2/a)$ on the trend record, $N_a \sim \text{Poisson}(\theta a)$ on the crash record. The tests whose acceptance region is an interval in T_a , randomized at most at its endpoints, form an essentially complete class: for every test there is an interval test of no larger size and of power no smaller at every alternative. Every admissible test is almost everywhere equal to an interval test.*

A.2 The acceptance region is the highest-density region

Lemma A.3 (Bridge to the highest-density region). *Fix an age a and a held model with predictive law for T_a . (i) If that law has a density $f_a^{\hat{\theta}}$ continuous at the realization, Ortoleva’s one-shot test at recording resolution δ —abandon when the realized cell has held-model probability at most $c\delta$ —converges as $\delta \downarrow 0$ to the density-cutoff rule: abandon iff $f_a^{\hat{\theta}}(T_a) \leq c$. (ii) Suppose every level set $\{f_a^{\hat{\theta}} = c\}$, $c > 0$, is Lebesgue-null—*

the density has no flat plateau. Then a cutoff $c_\alpha(a)$ calibrating $A_\alpha(a) = \{f_a^\theta \geq c_\alpha(a)\}$ of (4) to predictive mass exactly $1 - \alpha$ exists, and $A_\alpha(a)$ has minimum Lebesgue measure among all Borel sets of held-model mass at least $1 - \alpha$, uniquely so up to Lebesgue-null modification. (iii) For the atomic law (A.2), the same calibration yields the probability-ordered Sterne region of mass at least $1 - \alpha$, of minimum cardinality.

Proof. (i). Let $x = T_a$ and $C_\delta(x) = [x - \delta/2, x + \delta/2]$. By continuity of f_a^θ at x ,

$$\mathbb{P}^{\hat{\theta}}(T_a \in C_\delta(x)) = \int_{C_\delta(x)} f_a^\theta(y) dy = f_a^\theta(x) \delta (1 + o(1)) \quad (\delta \downarrow 0). \quad (\text{A.5})$$

The resolution- δ test abandons iff this cell probability is at most $c\delta$, i.e., by (A.5), iff $f_a^\theta(x)(1 + o(1)) \leq c$; letting $\delta \downarrow 0$ gives abandon whenever $f_a^\theta(x) < c$ and retain whenever $f_a^\theta(x) > c$. At the cutoff $f_a^\theta(x) = c$ the leading-order cell probability equals $c\delta$ and the test sits on its boundary; for the record densities of the main channels—the Gaussian trend statistic and other unimodal continuous laws, non-constant on the level set so it carries no Lebesgue mass— $\{f_a^\theta = c\}$ is null (a continuous density with a flat plateau at height c would be the exception, excluded here), so retaining iff $f_a^\theta(x) > c$ and iff $f_a^\theta(x) \geq c$ coincide, and we take the closed region $\{f_a^\theta \geq c\}$ of (4), fixing the $o(1)$ ambiguity; the atomic Sterne region (Lemma D.1) is closed by construction, the cutoff atom included to meet the mass floor.

(ii). With $f = f_a^\theta$ and $c = c_\alpha(a)$, the super-level set $A^* = \{f \geq c\}$ calibrated to mass $1 - \alpha$ has minimum Lebesgue measure among Borel sets of held-model mass at least $1 - \alpha$ —the highest-density-region property (Box and Tiao, 1973; Hyndman, 1996). (For the unimodal continuous record densities here $\{f = c\}$ is Lebesgue-null and a unique c attains mass exactly $1 - \alpha$.) Both steps are the generalized Neyman–Pearson lemma with Lebesgue measure— σ -finite, as the lemma requires—in the role of the alternative (Lehmann and Romano, 2005, Theorem 3.2.1); we display the

argument. For the exact calibration, $c \mapsto \mathbb{P}^{\hat{\theta}}(f_a^{\hat{\theta}} \geq c)$ is continuous—since $\mathbb{P}^{\hat{\theta}}(f_a^{\hat{\theta}} = c) = c \text{Leb}\{f_a^{\hat{\theta}} = c\} = 0$ —and decreases from 1 to 0, so $c_\alpha(a)$ exists. For uniqueness, let B have held-model mass at least $1 - \alpha$ and $\text{Leb}(B) = \text{Leb}(A_\alpha(a))$; abbreviate $A = A_\alpha(a)$, $c = c_\alpha(a)$. On $B \setminus A$ the density is strictly below c , on $A \setminus B$ it is at least c , and the mass constraint gives $\mathbb{P}^{\hat{\theta}}(A \setminus B) \leq \mathbb{P}^{\hat{\theta}}(B \setminus A)$; hence $c \text{Leb}(A \setminus B) \leq \mathbb{P}^{\hat{\theta}}(A \setminus B) \leq \mathbb{P}^{\hat{\theta}}(B \setminus A) < c \text{Leb}(B \setminus A)$ whenever $\text{Leb}(B \setminus A) > 0$, contradicting $\text{Leb}(B \setminus A) = \text{Leb}(A \setminus B)$; so both differences are Lebesgue-null.

(iii). For the atomic law (A.2), ordering the integers by descending mass (ties by smaller count) and taking the shortest initial segment to reach $1 - \alpha$ gives the probability-ordered Sterne region (Sterne, 1954), of minimum cardinality among sets of mass at least $1 - \alpha$: any set B of that mass with $|B| < |A|$ must omit some accrued atom $n \in A$ and include an atom $m \notin A$ with $p_m \leq p_n$ (accrual is by descending mass), so exchanging m for n raises the mass without raising cardinality; iterating, A 's initial segment of size $|B|$ already has mass at least that of B , yet by minimality of the accrual it has mass $< 1 - \alpha$, a contradiction. The boundary atom at the accrual cutoff is included—the closed Sterne interval of Lemma D.1, not the strict super-level set—so the region is uniquely defined. $\square$

Proposition 3 (The band from the primitive). *Fix an age a and a held model whose predictive law for T_a is unimodal—with, in the continuous channel, no level set of its density carrying mass; in the counting channel the atomic law of (A.2). Among Borel acceptance regions of held-model mass at least $1 - \alpha$: (i) exactness fixes the mass floor; (ii) every exact test is weakly dominated by an interval test (Lemma A.2); (iii) the fine-resolution limit of Ortoleva's cell rule is the density-cutoff region (Lemma A.3(i)), which, calibrated to the mass floor, is the unique minimal region up to null sets (Lemma A.3(ii))—an interval, by unimodality—and, in the counting channel, the*

probability-ordered Sterne region (Lemma A.3(iii)). The band of Definition 1 is thus pinned by the primitive under these regularity conditions, not by exactness alone: alternative exact-size selections—equal-tailed, unbiased—exist and are not generated by the primitive.

Proof. Parts (i)–(iii) are the cited lemmas. The one assembly step: under unimodality every superlevel set $\{f_a^{\hat{\theta}} \geq c\}$, $c > 0$, is an interval, so the minimal region of Lemma A.3(ii) lies *inside* the essentially complete class of Lemma A.2 rather than merely being dominated by a member of it, and the two lemmas select the same object. $\square$

A.3 What a trigger must test

Lemma A.4 (Ancillary triggers do not revise). *Under the re-selection rule (8): (i) the re-ranking depends on the record only through the stopped pair (T, τ) —the minimal sufficient statistic together with the regime length—so monitoring (T, τ) discards only menu-irrelevant detail; (ii) a re-ranking that consults the record only through a statistic ancillary for the menu (its law the same under every $\theta \in \Theta$) produces a null revision, $\rho' = \rho$ and $\hat{\theta}' = \hat{\theta}$; (iii) a belief-changing revision requires a re-ranking input that discriminates among menu models at the stopped pair—the statistic the re-selection (8) consults, not the band-exit trigger, which may itself be ancillary. The necessity is of menu-discrimination in the re-ranking input, not of the minimal sufficient statistic specifically. Discrimination is necessary but not sufficient: a model-changing re-selection also requires the induced likelihood ratios to overcome the current odds carried by ρ , and a model at zero current weight stays at zero.*

Proof. (i). By the Fisher–Neyman factorization (Lehmann and Casella, 1998, §1.6), sufficiency of the stopped pair (T, τ) for the menu—the statistic *and* the regime

length, the pair Assumption 1 certifies for stopped records—the likelihood at a stopping time of an exponential family of processes (Küchler and Sørensen, 1997)—(τ the regime *length*, the exposure since adoption, not a calendar date)—gives a factorization, with $g \geq 0$ free of θ ,

$$L_\theta(X_\tau) = g(X_\tau) k_\theta(T(X_\tau), \tau), \quad \theta \in \Theta. \quad (\text{A.6})$$

Substituting (A.6) into (8) and normalizing, the common factor g cancels:

$$\rho'(\theta) = \frac{L_\theta(X_\tau) \rho(\theta)}{\sum_{\theta''} L_{\theta''}(X_\tau) \rho(\theta'')} = \frac{k_\theta(T, \tau) \rho(\theta)}{\sum_{\theta''} k_{\theta''}(T, \tau) \rho(\theta'')}, \quad (\text{A.7})$$

so ρ' depends on the record only through the terminal pair (T, τ) .

(ii). Let S be ancillary for the menu, so its law—and hence its likelihood $L_\theta^S(s)$ —does not depend on θ : $L_\theta^S(s) = \bar{g}(s)$ for all θ . Re-ranking on S in place of the full record replaces $k_\theta(T)$ by $\bar{g}(S)$ in (A.7), giving

$$\rho'(\theta) = \frac{\bar{g}(S) \rho(\theta)}{\sum_{\theta''} \bar{g}(S) \rho(\theta'')} = \frac{\rho(\theta)}{\sum_{\theta''} \rho(\theta'')} = \rho(\theta), \quad (\text{A.8})$$

so $\rho' = \rho$ and $\hat{\theta}' = \arg \max \rho' = \arg \max \rho = \hat{\theta}$: the revision is null. The conclusion is about conditioning the re-ranking on S itself; a revision merely *triggered* by an ancillary event but *re-ranked* on the full record would still move ρ through the record's menu-discriminating part, so the null-revision property requires the ancillary statistic to be the re-ranking input, not just the trigger.

(iii). By (A.7), $\rho' \neq \rho$ for some realization—hence a possible change in $\arg \max$ —requires $\theta \mapsto k_\theta(T)$ to be non-constant, i.e. the *re-ranking input* T —not the band-exit trigger—must discriminate among menu models. Any menu-discriminating statistic suffices; the minimal sufficient statistic is monitored as a modeling economy (it dis-

cards only the ancillary factor g), not because no other statistic could revise. $\square$

A.4 The ranking accumulates the full history

Lemma A.5 (Chaining). *Let $0 = \sigma_0 < \sigma_1 < \dots < \sigma_k$ be successive revision times and apply (8) at each. Then for every pair $\theta, \theta_0 \in \Theta$ with $\rho_0(\theta_0) > 0$ (a model of zero initial mass keeps $\rho_t = 0$ under the multiplicative update and is dropped from the ratios),*

$$\frac{\rho_{\sigma_k}(\theta)}{\rho_{\sigma_k}(\theta_0)} = \frac{\rho_0(\theta)}{\rho_0(\theta_0)} \frac{d\mathbb{P}^\theta}{d\mathbb{P}^{\theta_0}} \Big|_{\mathcal{F}_{\sigma_k}}, \quad (\text{A.9})$$

so the ranking at σ_k is the single Bayes update of ρ_0 on the entire history up to σ_k . The held model is the full-history maximum-a-posteriori choice; only the test record restarts, the ranking ρ does not.

Proof. Let $\Lambda_t^{\theta/\theta_0} := d\mathbb{P}^\theta/d\mathbb{P}^{\theta_0} \Big|_{\mathcal{F}_t}$ be the density (likelihood-ratio) process, which by construction of the family of laws is consistent across t : for $s < t$,

$$\Lambda_t^{\theta/\theta_0} = \Lambda_s^{\theta/\theta_0} \cdot \frac{L_\theta(X_{(s,t]})}{L_{\theta_0}(X_{(s,t]})}, \quad (\text{A.10})$$

where $L_\theta(X_{(s,t]})$ denotes the likelihood of the increment *given* $\mathcal{F}_s$. The multiplicative property is the projectivity of Radon–Nikodym derivatives of consistent measures and holds for any process. At the revision times no optional-sampling step is needed: the σ_i are a.s. finite, on $\{\sigma_i < \infty\}$ the restriction of the locally equivalent laws to $\mathcal{F}_{\sigma_i}$ has density Λ_{σ_i} (Jacod and Shiryaev, 2003, §III.3), and in the canonical channels that density has the explicit exponential form (A.3)–(A.4), for which (A.10) at $s = \sigma_{i-1}$, $t = \sigma_i$ is an algebraic identity of the displayed densities, evaluated pathwise. For the independent-increment records of this paper—Poisson disaster counts and Gaussian trend increments—this conditional increment likelihood equals the *unconditional* law

of the segment, so the segment-likelihood form of $L_\theta(X_{(\sigma_{i-1}, \sigma_i]})$ is exact. For a general state-dependent record the factors remain conditional likelihoods given the past and the same telescoping holds in that conditional form; the unconditional-segment form is special to independent increments. Taking ratios in (8) at the i -th revision, applied to that regime's record $X_{(\sigma_{i-1}, \sigma_i]}$,

$$\frac{\rho_{\sigma_i}(\theta)}{\rho_{\sigma_i}(\theta_0)} = \frac{L_\theta(X_{(\sigma_{i-1}, \sigma_i]})}{L_{\theta_0}(X_{(\sigma_{i-1}, \sigma_i]})} \frac{\rho_{\sigma_{i-1}}(\theta)}{\rho_{\sigma_{i-1}}(\theta_0)}. \quad (\text{A.11})$$

Iterating (A.11) from $i = 1$ to k and telescoping the product of increment factors via (A.10),

$$\frac{\rho_{\sigma_k}(\theta)}{\rho_{\sigma_k}(\theta_0)} = \frac{\rho_0(\theta)}{\rho_0(\theta_0)} \prod_{i=1}^k \frac{L_\theta(X_{(\sigma_{i-1}, \sigma_i]})}{L_{\theta_0}(X_{(\sigma_{i-1}, \sigma_i]})} = \frac{\rho_0(\theta)}{\rho_0(\theta_0)} \Lambda_{\sigma_k}^{\theta/\theta_0}, \quad (\text{A.12})$$

which is (A.9). Since $\arg \max_\theta \rho_{\sigma_k}(\theta)$ depends only on the ratios (A.9), the held model is the full-history maximum-a-posteriori choice; the product is independent of how the history was segmented into regimes, so the timing of revisions does not affect the ranking, only the test record (4) is reset. $\square$

The lemma confirms the structural point of Section 2: the non-Bayesian feature is the maximum-a-posteriori selection together with the restart of the test record, not a reset of the second-order belief ρ , which accumulates full-history likelihood.

A.5 Restart is forced

Lemma A.6 (Restart removes an adoption-time degeneracy). *If a freshly adopted model were monitored against the inherited record rather than a restarted one, it could lie outside its own acceptance region at the instant of adoption, forcing an immediate revision with no new data. Restarting the test record at each adoption removes this degeneracy.*

Proof. We exhibit primitives in the crash channel (A.2) for which the degeneracy occurs. Take age $a = 1$ and tolerance $\alpha = 0.1$. For the incumbent rate $\lambda_2 = 10$, the held law of N_1 is $\text{Poisson}(10)$, whose Sterne region of mass at least 0.9 is, accruing integers by descending mass,

$$A_{\alpha}^{(\lambda_2)}(1) = \{5, 6, \dots, 15\}, \quad \mathbb{P}^{10}(5 \leq N_1 \leq 15) \approx 0.922 \geq 0.9, \quad (\text{A.13})$$

so its floor is $l^{(\lambda_2)} = 5$. Suppose the record is $N_1 = 3$: then $N_1 = 3 < 5$ is a lower exit from the λ_2 -band (a quiet spell), and the agent abandons λ_2 . The re-selection (8) re-ranks by the just-ended record, which as a quiet spell favours the lower rate: $L_{\lambda_1}(3) = e^{-1}/3! > L_{\lambda_2}(3) = e^{-10}10^3/3!$, a likelihood ratio $e^9/10^3 \approx 8.1$ for $\lambda_1 = 1$ over the incumbent. The carried ranking runs the other way— $\rho(\lambda_2) \geq \rho(\lambda_1)$, since λ_2 was the held model—so the front-runner tips to $\lambda_1 = 1$ exactly when that incumbency lead is milder than the record's factor; the full-history ranking (Lemma A.5) is what settles the contest when record and incumbency pull oppositely. This is the regime in which the adoption-time degeneracy can bite. The held law of N_1 under λ_1 is $\text{Poisson}(1)$, whose Sterne region of mass at least 0.9 is

$$A_{\alpha}^{(\lambda_1)}(1) = \{0, 1, 2\}, \quad \mathbb{P}^1(0 \leq N_1 \leq 2) = e^{-1}(1 + 1 + \tfrac{1}{2}) \approx 0.920 \geq 0.9, \quad (\text{A.14})$$

so its ceiling is $u^{(\lambda_1)} = 2$. The inherited record $N_1 = 3$ satisfies $3 > 2$, an *upper* exit from the λ_1 -band:

$$N_1 = 3 \notin A_{\alpha}^{(\lambda_1)}(1) = \{0, 1, 2\}. \quad (\text{A.15})$$

So testing the freshly adopted λ_1 against the inherited record rejects it at the instant of adoption, with no new data. Restarting the test record sets the new model's count to 0 at adoption, inside every band ($0 \in A_{\alpha}(a_0)$), and judges λ_1 only on

data accruing thereafter; the degeneracy is removed. This is the restart-after-alarm convention of sequential surveillance (Page, 1954). The exact Sterne endpoints used here are constructed in Section 2.4. $\square$

A.6 Probation onset

Lemma A.7 (Tempo / probation onset). (i) *In the crash channel (A.2), with $a_0 = 0$ the holding time satisfies $\tau > 0$ a.s.: continuous monitoring from adoption produces no instantaneous rejection.* (ii) *In the trend channel (A.1), when the held model is the truth, continuous monitoring from $a = 0$ rejects it at arbitrarily small ages, $\inf\{a > 0 : T_a \notin A_\alpha(a)\} = 0$ a.s.; a probation onset $a_0 > 0$ is therefore required for $\tau > 0$.*

Proof. (i). Let the held rate be λ and $m_\alpha := -\log(1 - \alpha)$. For small ages the held law $\text{Poisson}(\lambda a)$ puts mass $e^{-\lambda a}$ on $\{0\}$, and $e^{-\lambda a} \geq 1 - \alpha$ exactly for $a \leq a^\dagger := m_\alpha/\lambda$. The probability-ordered Sterne region is the single atom $\{0\}$ as long as 0 is also the modal point, $\lambda a \leq 1$; writing $\bar{a} := \min\{a^\dagger, 1/\lambda\} > 0$,

$$A_\alpha(a) = \{0\} \quad \text{for } 0 < a \leq \bar{a}. \quad (\text{A.16})$$

For $\alpha \leq 1 - e^{-1}$ one has $a^\dagger \leq 1/\lambda$ and $\bar{a} = a^\dagger$; for larger α , $\bar{a} = 1/\lambda$, a strictly smaller but still positive interval, which is all the conclusion $\tau > 0$ a.s. below requires. Let T_1 be the first jump time of N ($T_1 \sim \text{Exp}(\lambda)$ under the held law, $\text{Exp}(\lambda^*)$ under the physical law; only $T_1 > 0$ a.s. is used). For $0 < a < T_1 \wedge \bar{a}$ we have $N_a = 0 \in A_\alpha(a)$ by (A.16) (the count is still zero strictly before the first jump), hence no exit on that open range, and therefore

$$\tau \geq T_1 \wedge \bar{a} > 0 \quad \text{a.s.} \quad (\text{A.17})$$

So $a_0 = 0$ is admissible: there is no a.s. instantaneous rejection, and an early exit (a disaster before $a^\dagger$) is a genuine surprise, not an artifact of continuous monitoring.

(ii). Let the held drift equal the truth, $\hat{\mu} = \mu^*$. Then under $\mathbb{P}^*$, $X_a - \mu^*a = \sigma W_a$, so the standardized statistic is

$$\frac{\bar{X}_a - \hat{\mu}}{\sigma/\sqrt{a}} = \frac{(X_a - \mu^*a)/a}{\sigma/\sqrt{a}} = \frac{\sigma W_a/a}{\sigma/\sqrt{a}} = \frac{W_a}{\sqrt{a}}, \quad (\text{A.18})$$

and by (6) the band $\{|\bar{X}_a - \hat{\mu}| \leq z_\alpha \sigma/\sqrt{a}\}$ is the event $\{|W_a|/\sqrt{a} \leq z_\alpha\}$. The small-time law of the iterated logarithm for Brownian motion holds: applying the law of the iterated logarithm at infinity (Khinchine, 1924) to the time-inverted Brownian motion $\widetilde{W}_s := sW_{1/s}$ and substituting $s = 1/a$ gives

$$\limsup_{a \downarrow 0} \frac{|W_a|}{\sqrt{2a \log \log(1/a)}} = 1 \quad \text{a.s.} \quad (\text{A.19})$$

Fix a sample path outside the null set where (A.19) fails, and along it choose a path-dependent sequence $a_n \downarrow 0$ realizing the lim sup, so that $|W_{a_n}|/\sqrt{2a_n \log \log(1/a_n)} \rightarrow 1$. Then

$$\frac{|W_{a_n}|}{\sqrt{a_n}} = \frac{|W_{a_n}|}{\sqrt{2a_n \log \log(1/a_n)}} \cdot \sqrt{2 \log \log(1/a_n)} \rightarrow +\infty, \quad (\text{A.20})$$

the first factor tending to 1 and the second to ∞ —a statement along the sequence, not a manipulation of the lim sup. So $|W_{a_n}|/\sqrt{a_n} > z_\alpha$ for all large n , and since the excluded null set does not depend on the path, this gives

$$\inf\{a > 0 : |W_a|/\sqrt{a} > z_\alpha\} = 0 \quad \text{a.s.}, \quad (\text{A.21})$$

i.e. the truth is rejected at arbitrarily small ages. Imposing a *fixed* probation onset $a_0 > 0$ in (5) replaces the infimum by $\inf\{a > a_0 : \dots\}$, which is positive, and

excludes the small-time pathology. The onset a_0 is a recording-environment primitive held fixed as $\delta \downarrow 0$, so the continuous-time limit is nondegenerate. $\square$

A.7 The lifetime reading consults the menu

Lemma A.8 (Time-uniform reading). *Let θ_0 be the held model and $\{\omega_\theta\}_{\theta \neq \theta_0}$ weights with $\omega_\theta \geq 0$, $\sum \omega_\theta = 1$. The rule that abandons θ_0 the first time the menu likelihood-ratio process*

$$\Lambda_t := \sum_{\theta \neq \theta_0} \omega_\theta \frac{d\mathbb{P}^\theta}{d\mathbb{P}^{\theta_0}} \Big|_{\mathcal{F}_t} \quad (\text{A.22})$$

reaches $1/\alpha$ has lifetime type-I error at most α : the probability of ever abandoning a correct θ_0 is at most α . This rule consults the alternatives. By the admissibility of anytime-valid inference, any test with such a lifetime guarantee is of nonnegative-supermartingale (e-process) form; the finite-menu likelihood-ratio mixture (A.22) is one such supermartingale, sufficient for the guarantee but not the only one—conservative rules and valid supermartingales not expressible as a mixture over the menu alternatives also control the lifetime error, and a supermartingale need not be a likelihood ratio to any named menu rate. We cite the supermartingale characterization and do not rely on it.

Proof. Each density process $d\mathbb{P}^\theta/d\mathbb{P}^{\theta_0} \Big|_{\mathcal{F}_t}$ is a nonnegative $\mathbb{P}^{\theta_0}$ -martingale with mean 1 and initial value 1 (Lemma A.1). A nonnegative convex combination of them, Λ in (A.22), is therefore a nonnegative $\mathbb{P}^{\theta_0}$ -martingale with

$$\Lambda_0 = \sum_{\theta \neq \theta_0} \omega_\theta \cdot 1 = 1, \quad \mathbb{E}^{\theta_0}[\Lambda_t] = 1. \quad (\text{A.23})$$

Ville’s inequality (Ville, 1939)—the maximal inequality stating that for a nonnegative supermartingale Λ with $\Lambda_0 = 1$ and any level $\beta \geq 1$, $\mathbb{P}^{\theta_0}(\sup_{t \geq 0} \Lambda_t \geq \beta) \leq 1/\beta$ —

applied with $\beta = 1/\alpha$ gives

$$\mathbb{P}^{\theta_0}\left(\sup_{t \geq 0} \Lambda_t \geq 1/\alpha\right) \leq \alpha. \quad (\text{A.24})$$

The lifetime test abandons θ_0 at $\inf\{t : \Lambda_t \geq 1/\alpha\}$, an event contained in $\{\sup_t \Lambda_t \geq 1/\alpha\}$, so by (A.24) the probability of ever abandoning a correct θ_0 is at most α . The statistic Λ in (A.22) involves the densities of the alternatives, so the lifetime rule consults the menu. The converse—that any lifetime (time-uniform) guarantee must rest on a nonnegative supermartingale with unit initial value—an e-process, the likelihood ratio being the canonical instance—is the admissibility characterization of Ramdas et al. (2020) and Grünwald et al. (2024); we invoke it only to motivate the per-test/lifetime dichotomy of §2.1, and the baseline rule of Definition 1 does not depend on it. $\square$

The per-test reading is not merely the literal transcription of Ortoleva’s rule into continuous time; it has a per-shift size-control rationale. Full dynamic consistency is not on the table: Epstein and Le Breton (1993) show that conditional beliefs reconciling every conditioning event—dynamically consistent in the strong sense—must be a single prior updated by Bayes’ rule, which leaves no room for the discrete model-switching we study. A rule that abandons one model for another *by surprise* is therefore non-Bayesian and necessarily forgoes that consistency; the switch *is* the inconsistency, and what the per-test design controls is its size. Weinstein (2017) studies Bayesian updating *tempered* by a classical size- α test, gating each non-Bayesian paradigm shift behind a size- α rejection, and shows that the tempered agent is approximately dynamically consistent, the departures confined to probability α per test. Our continuous-time rule is the $\delta \downarrow 0$ limit of exactly that gate. The recurrence we dwell on (Section 2.4)—a correct model abandoned infinitely often—is that confined

inconsistency compounded over a lifetime: the multiple-testing cost of the per-test design, kept as the engine of belief dynamics rather than controlled away.

Proof of Lemma A.2. At fixed age a the record law in either channel is a full one-parameter exponential family with T_a the sufficient statistic: Gaussian location (known variance) on the trend record, Poisson on the crash record. For testing a point null against the two-sided alternative in such a family, the tests with convex acceptance regions in the sufficient statistic are essentially complete, and the acceptance region of every admissible test is convex up to a null set (Birnbaum, 1955; Matthes and Truax, 1967). In one dimension a convex region is an interval; on a lattice support its boundary carries atoms, so attaining a prescribed size can require randomizing there, and never elsewhere. The Gaussian boundary is null, so no randomization arises in the trend channel; in the crash channel exact size α generally requires boundary randomization, and the nonrandomized probability-ordered interval of held-model mass at least $1 - \alpha$ in (7) is the conservative selection. The unbiased selection is the classical one (Lehmann and Romano, 2005, §4.2). $\square$

B The surviving record as a state variable: proof

Proof of Theorem 1. Part (i): insufficiency. The revision time under any rule of Definition 1 is the first failure of the retention test on the since-adoption record $X_{(\sigma,t]}$, a functional of the adoption date σ and the public path; a fixed prior updated on $\mathcal{F}_t$ is a functional π_t of the public path alone, identical across adoption dates. We exhibit an event of positive probability on which two holders of the same $\hat{\theta}$ share π_t yet carry different conditional revision intensities; the revision law then cannot be $\sigma(\pi_t)$ -measurable.

Work in the crash channel at held rate $\hat{\lambda}$ with band $[l(a), u(a)]$ (Lemma D.1).

Within an age. Fix an age a with $u(a) - l(a) \geq 2$, off the finitely many step ages of the band (such ages exist: the band (A.13) has width ten). Off the step ages, $u(a + da) = u(a)$ for all small da ; one arrival moves a ceiling-holder to $u(a) + 1 > u(a)$ and a slack-two holder to at most $u(a) - 1$, and two arrivals in $(a, a + da]$ have probability $O(da^2)$, so

$$\mathbb{P}^{\lambda^*}(N_{a+da} > u(a+da) \mid N_a = u(a)) = \lambda^* da + o(da), \quad \mathbb{P}^{\lambda^*}(N_{a+da} > u(a+da) \mid l(a) \leq N_a \leq u(a)-2) = 0, \quad (\text{B.1})$$

the held model is insufficient within an age. *Across ages on one history.* Take $\hat{\lambda} = \lambda^* = 10$ and $\alpha = 0.1$, adoption dates $\sigma_A < \sigma_B$ with ages $a_A = 1$ and $a_B = \frac{1}{2}$ at t (chosen off both bands' step ages, as we may): the bands are $A_\alpha(1) = \{5, \dots, 15\}$ of mass ≈ 0.922 (A.13) and $A_\alpha(\frac{1}{2}) = \{2, \dots, 9\}$, the Sterne region of Poisson(5), of mass $\approx 0.928 \geq 0.9$. The records are counts over the nested windows $(\sigma_A, t] \supset (\sigma_B, t]$, whose two halves are independent Poisson(5) increments, so the event

$$E = \{N_{(\sigma_B, t]} = 7\} \cap \{N_{(\sigma_A, \sigma_B]} = 8\}, \quad \mathbb{P}^{\lambda^*}(E) = e^{-5} \frac{5^7}{7!} \cdot e^{-5} \frac{5^8}{8!} > 0, \quad (\text{B.2})$$

puts holder B at $N_{a_B} = 7$ with $l(\frac{1}{2}) = 2 \leq 7 \leq u(\frac{1}{2}) - 2 = 7$, strictly inside her band, and holder A at $N_{a_A} = 7 + 8 = 15 = u(1)$, the ceiling of his. The endpoint counts alone do not yet make A and B *current* holders at t —some paths in E exit a band earlier—so refine E by an in-band staircase: on the finite grid formed by both holders' scheduled band-step ages in $(\sigma_A, t]$, prescribe a nondecreasing integer path lying in the ceiling-lagged, floor-led bands of both windows at every grid age, with the endpoint values of E . Such a path exists because each floor reaches a given integer only after the corresponding ceiling has passed it (the staircase of Lemma D.1), and the refined event $E' \subseteq E$ prescribes finitely many Poisson increments over disjoint intervals, each

of positive probability, so $\mathbb{P}^{\lambda^*}(E') > 0$; on E' no earlier exit occurs, the grid conditions forbidding crossings at real ages exactly as in (D.18). Two scopings make the excluded object exact. The σ -field is $\sigma(\pi_t)$ —functionals of the public path under the fixed prior: the witness exhibits two admissible full states on one path with the same π_t and different exit intensities, so no $\sigma(\pi_t)$ -measurable description carries the hazard. And the adoption dates $\sigma_A < \sigma_B$ are *initialized* states of the canonical construction—initial conditions of the flow, not claimed as a positive-probability configuration of a single rule-generated path from a common prior; adjoining the private adoption date, the supplementary variable of part (i), is exactly what restores the Markov description—the theorem’s point, not its concession. Both hold $\hat{\lambda}$; both observe the same public path, so π_t is the same random variable for both; by (B.1) their upper-exit intensities at t are 0 and λ^* . Since $\mathbb{P}^{\lambda^*}(E') > 0$, the conditional revision law is not $\sigma(\pi_t)$ -measurable; what separates the holders is the private adoption date σ , equivalently the age $a = t - \sigma$, which π_t does not carry. (The integer endpoints can coincide over age ranges, so the witness is the event E' , not $a_A \neq a_B$ alone.) The argument used only reset-on-adoption against the posterior’s reset-freeness, so it holds for the whole record-resetting class.

Part (ii): irreducibility. The revision time is the first exit of T_a from $A_a(a)$; the band is fixed by $(\hat{\theta}, a)$ and the data drive T forward given (T_a, a) , so the revision law is a measurable function of $(\hat{\theta}, T_a, a)$. No coordinate drops: T_a *given* $(\hat{\theta}, a)$, the within-age pair of Part (i); a *given* $(\hat{\theta}, T_a)$, the band moves with age—stepping in the crash channel, contracting like $a^{-1/2}$ in the trend channel: a summary on the age- a edge, $|T - \hat{\mu}| = z_\alpha \sigma / \sqrt{a}$, is interior at any earlier age $a' < a$, since $z_\alpha \sigma / \sqrt{a} < z_\alpha \sigma / \sqrt{a'}$ by (6); $\hat{\theta}$ *given* (T_a, a) , distinct held models impose bands that separate: trend-channel bands have distinct centers $\hat{\mu}_1 \neq \hat{\mu}_2$, and crash-channel ceilings separate linearly in

age,

$$u_{\hat{\lambda}_1}(a) - u_{\hat{\lambda}_2}(a) = (\hat{\lambda}_1 - \hat{\lambda}_2) a + O(\sqrt{a}) \longrightarrow \infty \quad (\hat{\lambda}_1 > \hat{\lambda}_2)$$

by the leading form of Lemma D.1, so the band functions differ from some age on even where the integer endpoints coincide at isolated ages. This uses age-moving bands and model-specific separation, so it holds in the two channels and in any with a nondegenerate age-dependent band and a menu-discriminating statistic, not by record-resetting alone. The comparison underlying non-measurability is between *admissible* full states sharing (T_a, a) : with distinct held models the continuation first-exit laws differ from the separation age on, and such same-record two-holder configurations are admissible as initialized states of the canonical construction, and the overlap domain is nonempty: at $\hat{\lambda}_1 = 10$, $\hat{\lambda}_2 = 9$, $\alpha = 0.1$, age 1, the bands are $A_{0.1}^{(10)}(1) = \{5, \dots, 15\}$ (A.13) and $A_{0.1}^{(9)}(1) = \{4, \dots, 13\}$, the Sterne region of Poisson(9) of mass $\approx 0.905 \geq 0.9$, so every count $n \in \{5, \dots, 13\}$ lies in both, and two initialized age-1 holders at the same $(n, 1)$ carrying the two models are admissible states whose continuation first-exit laws differ from the separation age on. On sub-domains where the bands are disjoint, (T_a, a) identifies $\hat{\theta}$ and non-measurability there is vacuous relabeling; the claim is irreducibility on the overlap domain, occupied under any initialization charging both models. States outside the held band exit immediately by convention and are excluded from the domain.

Part (iii): closure. Between revisions $\hat{\theta}$ and ρ are constant, the age flows ($\dot{a} = 1$), and T_a evolves by the data's Markov transition given (T_a, a) ; at the first exit the state jumps and the record resets, $a \rightarrow 0$. By the stopped sufficiency of Assumption 1 the re-ranking at τ depends on the record only through (T_τ, τ) , so ρ' and $\hat{\theta}' = \arg \max \rho'$ are functions of (ρ, T_τ, τ) . Deterministic flow between jumps, a first-exit jump time, and a state-function reset are the transitions of a piecewise-deterministic Markov process

(Davis, 1984) in the crash channel and of a boundary-reset diffusion in the trend channel; with the data Markov the enlarged state is therefore Markov (Küchler and Sørensen, 1998), revision its first exit from (4). It does not regress to a finer object: ρ is a finite sufficient statistic for the history’s effect on the ranking (Appendix A), and $(\hat{\theta}, T_a, a)$ alone suffices for the revision law, ρ entering only the post-revision adoption. All claims are over a positive time interval, never by conditioning on an instantaneous event (§2.3). $\square$

Adaptedness versus factorization. For a single agent with fixed initialization σ_t is itself $\mathcal{F}_t$ -measurable—the last exit time, a functional of the data—so the revision time and the compensator of the revision point process are $\mathcal{F}_t$ -adapted. The non-representability is therefore not a failure of adaptedness but of *factorization*: the revision law is $\sigma(\hat{\theta}_t, T_{a_t}, a_t)$ -measurable but not $\sigma(\beta_t)$ -measurable, the two σ -algebras being in general incomparable—the posterior carries the menu weights, the record carries the age. (A literal filtration enlargement appears only under randomized initialization, when the same fact appears as the revision law’s projection onto the posterior-generated subfiltration being a coarsened average.)

Realized, not merely constructed. Proposition 1(b) turns the held-model half of the claim from a constructed possibility into a path regularity: along almost every trajectory the model is re-adopted infinitely often at a range of ages, so the same held belief recurs paired with distinct hazards on the agent’s own path. The stronger statement—that no fixed-prior posterior suffices—remains the construction’s, comparing two agents at one date rather than one agent across dates.

C The revision hazard as a measure

Section 2.3 derives the hazard from the first-exit law and states the marginal/conditional distinction; this appendix records the hazard measure and its channel split. Throughout, the investor's type $(\hat{\theta}, \alpha)$ is fixed: every object below is a fibre object at that type, the tolerance entering through the band endpoints u_α, l_α , and we suppress the type in H , S , and $h(a)$ only where the fibre is unambiguous—no cross-type average appears in this appendix. In the disputed-crash-rate channel the band endpoints (7) step at discrete ages (Lemma D.1), so revision carries point masses and the full object is the generalized hazard measure

$$H(da) = h(a) da + \sum_j p_j \delta_{t_j}(da), \quad p_j = \mathbb{P}(\tau = t_j \mid \tau > t_j^-), \quad (\text{C.1})$$

with $\{t_j\}$ the deterministic band-step dates, and model-retention survival the product integral

$$S(a) = \exp\left(-\int_0^a h(u) du\right) \prod_{t_j \leq a} (1 - p_j), \quad \Delta S(t_j) = -p_j S(t_j^-). \quad (\text{C.2})$$

The marginal hazard splits by channel. *Crash channel:* the inaccessible upper exits carry a genuine state-dependent jump rate $\chi(\hat{\theta}, x, a; \alpha) = \lambda^* \mathbf{1}\{x = u_\alpha(a)\}$ —the next disaster is the exit only when the count already sits on the ceiling—and the absolutely continuous fibre hazard $h(\hat{\theta}, \alpha, a)$ is the average of this rate over the survivor law $\nu_a(\cdot \mid \hat{\theta}, \alpha, a)$ of the record position,

$$h(\hat{\theta}, \alpha, a) = \int \chi(\hat{\theta}, x, a; \alpha) \nu_a(dx \mid \hat{\theta}, a) = \lambda^* \nu_a(\{u_\alpha(a)\} \mid \hat{\theta}, a), \quad (\text{C.3})$$

an ordinary integral precisely because χ is a rate. *Trend channel:* the exit is a

predictable boundary hit with no absolutely continuous rate, so there is no position-conditional rate to fibre-average; the marginal holding-time hazard, where it exists, is generated by boundary *flux*—the rate at which ν_a leaks across the moving edge—not by integrating a finite χ . We use χ in the trend channel only as a proximity heuristic, never as a rate entering (C.3). The compensator of the revision point process is accordingly mixed: absolutely continuous only where exits are inaccessible (the crash channel’s disaster-triggered upper exits), singular where predictable (the trend boundary and the scheduled floor steps). So the fibre hazard $h(\hat{\theta}, \alpha, a)$ is recovered from the holding-time *law* (9)—the marginal age- a revision rate of the type, an average over surviving record positions—while conditional on the full enlarged state the operative objects are the crash-channel rate $\chi(\hat{\theta}, x, a; \alpha)$ together with the scheduled atoms and the trend boundary flux; it is not the survival-conditional expectation of a pathwise intensity, which in the trend channel does not exist. We keep the distributional measure H and the pathwise first-exit distinct throughout; (C.1) is well defined as a measure, and we claim no closed form for h in general.

D The life of a held model: proofs

This appendix proves the results of Section 2.4. Throughout, the held model is $\hat{\theta}$ (a drift $\hat{\mu}$ or a rate $\hat{\lambda}$), the truth is $\mathbb{P}^*$ (drift μ^* or rate λ^*), and the band is (6)–(7). We first record the structure of the crash-channel band, used in Proposition 1(b) and Lemma D.3.

Lemma D.1 (Sterne band structure). *In the disputed-crash-rate channel the band endpoints $l_\alpha(a), u_\alpha(a)$ of (7) are deterministic step functions of age—piecewise constant with finitely many jumps on any compact interval, taken right-continuous by convention at the step ages (a choice that fixes the band only at those finitely many*

ages; under an infimum definition of the first-exit time it leaves the holding-time law unchanged, but it is not immaterial to membership at the exit time—a crash-channel lower exit occurs exactly when the deterministic floor steps up past an unchanged integer count, i.e. at a step age—and to the bookkeeping of predictable atoms, whose dates are these floor-steps—and nested in tolerance: $A_{\alpha'}(a) \subseteq A_{\alpha}(a)$ for $\alpha' > \alpha$ at every a . As $a \rightarrow \infty$ the endpoints obey the normal approximation $l_{\alpha}(a), u_{\alpha}(a) = \hat{\lambda}a \mp z_{\alpha}\sqrt{\hat{\lambda}a} + o(\sqrt{a})$, a half-width $z_{\alpha}\sqrt{\hat{\lambda}a} = O(\sqrt{a})$ about the held mean $\hat{\lambda}a$.

Proof. The held law of N_a is $\text{Poisson}(\hat{\lambda}a)$ with masses $p_n(a) = e^{-\hat{\lambda}a}(\hat{\lambda}a)^n/n!$. The Sterne region accrues integers in order of descending $p_n(a)$ until the cumulative mass first reaches $1 - \alpha$. The region changes with a only when either two integers swap mass rank, $p_n(a) = p_m(a)$ for some $n \neq m$, or the accrued mass crosses $1 - \alpha$ at a new integer. The rank-swap condition is

$$p_n(a) = p_m(a), \ n < m, \ a > 0 \iff (\hat{\lambda}a)^{m-n} = \frac{m!}{n!} \iff a = \frac{1}{\hat{\lambda}} \left(\frac{m!}{n!} \right)^{1/(m-n)}, \quad (\text{D.1})$$

a single age for each pair, where the equivalence divides by $(\hat{\lambda}a)^n$ and so requires $a > 0$; at $a = 0$ the held law is the point mass $p_0 = 1$, with no rank-swap. On a compact interval $[0, A]$ the band lies in $\{0, \dots, N_A^{\max}\}$ with $N_A^{\max}$ finite: writing $U_A := \min\{k : \mathbb{P}^{\text{Poisson}(\hat{\lambda}A)}(N \leq k) \geq 1 - \alpha\}$, the benchmark set $\{0, \dots, U_A\}$ has mass at least $1 - \alpha$ under $\text{Poisson}(\hat{\lambda}a)$ for every $a \leq A$ (the Poisson family is stochastically increasing in its mean), so the minimal-cardinality region has at most $U_A + 1$ atoms; it contains the modal integer $\lfloor \hat{\lambda}a \rfloor \leq \hat{\lambda}A$, where the accrual starts, so its ceiling is at most $N_A^{\max} := \lfloor \hat{\lambda}A \rfloor + U_A + 1$. Hence only finitely many pairs (n, m) and finitely many rank-swap ages (D.1) arise; between consecutive rank-swap ages the descending-mass *ordering* is constant. With the ordering fixed the region can still

change at a *cumulative-threshold* age, where the accrued mass of the leading segment first reaches $1 - \alpha$ at a new integer. But for a fixed ordering the cumulative mass of any leading segment is a finite sum of terms $e^{-\hat{\lambda}a}(\hat{\lambda}a)^n/n!$, real-analytic and non-constant in a , so it equals $1 - \alpha$ at finitely many ages on $[0, A]$; finitely many cumulative-threshold ages thus arise per ordering, and finitely many in all. Between consecutive ages of either kind the region is constant. Right-continuity is by the convention of taking the region at a : at a cumulative-mass threshold age, where the minimal Sterne region jumps, the post-step (larger) region is in force from the step age onward—the convention the crash-channel lower-exit and the holding-time transfer matrix below both use, so that a scheduled floor or ceiling step takes effect *at* its date rather than just after it. The convention is a tie-break confined to the finitely many exact-attainment ages, where two probability-ordered regions of mass at least $1 - \alpha$ coexist—the minimal one and its one-integer enlargement; the right-continuous choice takes the enlargement at the date, minimal cardinality holding at every other age, so the bookkeeping at the predictable step dates and Sterne minimality never conflict off this finite set. Formally two objects coexist: the fixed-age minimal region of Definition 1 and the right-continuous *process* version the dynamics use, equal off the finite exact-attainment set and differing there by the enlargement; the lower-exit atoms and the transfer matrices below use the process version. For nestedness, $\alpha' > \alpha$ gives a smaller target mass $1 - \alpha' < 1 - \alpha$; since both regions accrue integers in the same descending-mass order, the α' -region is an initial segment of the α -region, so $A_{\alpha'}(a) \subseteq A_{\alpha}(a)$ (with equality when the two targets fall within the same mass step). For the asymptotic half-width, the highest-density region of $\text{Poisson}(\hat{\lambda}a)$ of mass $1 - \alpha$ converges, under the normal approximation $\text{Poisson}(\hat{\lambda}a) \approx \mathcal{N}(\hat{\lambda}a, \hat{\lambda}a)$ valid as $\hat{\lambda}a \rightarrow \infty$ by the local central limit theorem for lattice laws (Petrov, 1975, Ch. VII), to the Gaussian highest-density interval $\hat{\lambda}a \pm z_{\alpha}\sqrt{\hat{\lambda}a}$, giving $l_{\alpha}(a), u_{\alpha}(a) = \hat{\lambda}a \mp z_{\alpha}\sqrt{\hat{\lambda}a} + o(\sqrt{a})$. $\square$

D.1 Non-explosion of revisions

Lemma D.2 (Non-explosion of revisions). *Run the per-test rule of Definition 1 in either canonical channel at a fixed tolerance, restarting the record at each adoption. The revision times satisfy $\sigma_k \uparrow \infty$ $\mathbb{P}^*$ -almost surely—the regimes do not accumulate at a finite time. The argument uses only a per-channel lower-tail floor on holding times—deterministic ($\tau_k \geq a_0$) in the trend channel, a positive lower-tail probability in the crash channel (Lemma A.7)—and the finiteness of the menu; it invokes neither consolidation nor the finiteness of any individual τ_k .*

Proof. Each σ_k is a stopping time, and the fundamental has independent increments (Brownian-with-drift in the trend channel, Poisson in the crash channel), so it is strong Markov at σ_k ; with the record restarted at adoption (Lemma A.6) the conditional law of $\tau_k = \sigma_{k+1} - \sigma_k$ given $\mathcal{F}_{\sigma_k}$ depends only on the post-revision held model $\hat{\theta}_{\sigma_k}$, which ranges over the *finite* menu Θ ; the data are time-homogeneous, so this conditional law is one of finitely many fresh-start laws. Each carries a lower-tail floor (Lemma A.7): $\tau_k \geq a_0$ surely in the trend channel, and $\mathbb{P}(\tau_k \geq \delta \mid \mathcal{F}_{\sigma_k}) \geq e^{-\lambda^* \delta}$ in the crash channel for all small δ (the wait to the first disaster off the singleton band $\{0\}$). Hence there are $\delta > 0$ and $q \in (0, 1]$ with $\mathbb{P}(\tau_k \geq \delta \mid \mathcal{F}_{\sigma_k}) \geq q$ for every k , so $\sum_k \mathbb{P}(\tau_k \geq \delta \mid \mathcal{F}_{\sigma_k}) \geq \sum_k q = \infty$ a.s., and Lévy’s conditional Borel–Cantelli lemma (Williams, 1991, §12.15) gives $\tau_k \geq \delta$ for infinitely many k a.s. Therefore $\sum_k \tau_k = \infty$ and $\sigma_k \uparrow \infty$ a.s. The argument invokes only the floor and the finite menu—not the finiteness of any individual τ_k , nor consolidation—so it is available wherever non-explosion is needed below, with no circularity. $\square$

D.2 Consolidation

Proof of Proposition 1, part (a). Work within $\text{supp}(\rho_0)$: a model with $\rho_0(\theta) = 0$ keeps $\rho_t(\theta) = 0$ under the multiplicative update and never leads, and $\theta^\circ \in \text{supp}(\rho_0)$ since $\rho_0(\theta^\circ) > 0$. By the chaining identity (Lemma A.5), the re-ranking at any revision equals re-ranking on the entire history to that date; in pairwise-ratio form, for $\theta, \theta^\circ \in \text{supp}(\rho_0)$ the ranking ratio carried at a revision time t is

$$\log \frac{\rho_t(\theta)}{\rho_t(\theta^\circ)} = \log \frac{\rho_0(\theta)}{\rho_0(\theta^\circ)} + \log \Lambda_t^{\theta/\theta^\circ}, \quad \Lambda_t^{\theta/\theta^\circ} := \frac{d\mathbb{P}^\theta}{d\mathbb{P}^{\theta^\circ}} \Big|_{\mathcal{F}_t}, \quad (\text{D.2})$$

the full-history likelihood ratio, the normalizing constant having cancelled. Between revisions ρ is frozen (Definition 1); the object that evolves continuously is the latent ratio $\Lambda_t^{\theta/\theta^\circ}$, and the law of large numbers below is a statement about it. Under each menu law the record has *stationary independent increments* (Brownian motion with drift, or a Poisson process), so the log-likelihood-ratio process $t \mapsto \log \Lambda_t^{\theta/\theta^\circ}$ is, under $\mathbb{P}^\star$, an additive functional with stationary independent increments. Write $\kappa(\theta) := \lim_{t \rightarrow \infty} \frac{1}{t} \text{KL}(\mathbb{P}_t^\star \| \mathbb{P}_t^\theta)$ for the per-unit-time relative entropy *rate*—the finite-horizon divergence over horizon length, the finite object here, since the full-path laws are mutually singular and the infinite-horizon divergence is not finite. The one-unit log-likelihood-ratio increment has finite first moment under $\mathbb{P}^\star$ in both channels—it is Gaussian on the trend record and an affine function of a Poisson count on the crash record, by (A.3)–(A.4)—so the rates $\kappa(\theta), \kappa(\theta^\circ)$ are finite. Then the drift per unit time is

$$\mathbb{E}^\star[d \log \Lambda_t^{\theta/\theta^\circ}] = (\kappa(\theta^\circ) - \kappa(\theta)) dt. \quad (\text{D.3})$$

By the strong law of large numbers for processes with stationary independent increments with integrable unit increment (Sato, 1999, Thm. 36.5)—applied directly to

calendar time rather than to a fixed grid—

$$\frac{1}{t} \log \Lambda_t^{\theta/\theta^\circ} \xrightarrow{\text{a.s.}} \kappa(\theta^\circ) - \kappa(\theta) \quad (t \rightarrow \infty). \quad (\text{D.4})$$

By the uniqueness of θ° as the Kullback–Leibler minimizer (in rate), $\kappa(\theta) > \kappa(\theta^\circ)$ for $\theta \neq \theta^\circ$, so the limit (D.4) is strictly negative; this is the continuous-time form of the concentration of the misspecified posterior on the KL projection (Berk, 1966). Hence $\log \Lambda_t^{\theta/\theta^\circ} \rightarrow -\infty$ a.s. for every $\theta \neq \theta^\circ$ in the (finite) support. This is a statement about the latent ratio in continuous *calendar* time, and it invokes no revisions. For each $\theta \neq \theta^\circ$ there is therefore an a.s. finite time after which $\log(\rho_0(\theta)/\rho_0(\theta^\circ)) + \log \Lambda_t^{\theta/\theta^\circ} < 0$; taking the maximum over the finite support, there is an a.s. finite calendar time T° such that, for every $t \geq T^\circ$ and every $\theta \neq \theta^\circ$,

$$\log \frac{\rho_0(\theta)}{\rho_0(\theta^\circ)} + \log \Lambda_t^{\theta/\theta^\circ} < 0. \quad (\text{D.5})$$

This is an inequality for the *latent* full-history odds; the carried ranking ρ is frozen between revisions, and equals the latent odds exactly at every revision time (Lemma A.5), so (D.5) is transferred to ρ at the revision times and only there. Hence at *any* revision occurring at a time $\sigma_k \geq T^\circ$, $\arg \max_\theta \rho_{\sigma_k}(\theta) = \theta^\circ$ a.s.—the lemma’s conditional implication, which asserts nothing about whether such a revision occurs (that is supplied by part (b); non-explosion, Lemma D.2, uses only the holding-time floor of Lemma A.7 and the finite menu, not part (a)). When the truth is on the menu, θ° is the truth, the unique zero of $\text{KL}(\mathbb{P}^* \|\cdot)$. Positive prior mass on the consolidation target is necessary— $\rho_0(\theta^\circ) > 0$, not full support: if $\rho_0(\theta^\circ) = 0$ then $\rho_t(\theta^\circ) = 0$ for all t by the multiplicative update, and θ° is never selected; absent the truth from the support, consolidation is to the KL-closest model within $\text{supp}(\rho_0)$. $\square$

D.3 Recurrence

Proof of Proposition 1, part (b). Trend channel, correct held model ($\hat{\mu} = \mu^\star$). By (A.18) the band is the event $\{|W_a|/\sqrt{a} \leq z_\alpha\}$, i.e. Brownian motion against the square-root boundary $z_\alpha\sqrt{a}$, the first-exit problem of Breiman (1967). Put $\tau' = \log a$ and

$$V_{\tau'} := e^{-\tau'/2} W_{e^{\tau'}}, \quad \text{so that} \quad R_a = \frac{W_a}{\sqrt{a}} = V_{\log a}. \quad (\text{D.6})$$

The time-changed process $M_{\tau'} := W_{e^{\tau'}}$ has quadratic variation $d\langle M \rangle_{\tau'} = d(e^{\tau'}) = e^{\tau'} d\tau'$, so $dM_{\tau'} = e^{\tau'/2} dB_{\tau'}$ for a standard Brownian motion B . By Itô's product rule applied to (D.6),

$$dV_{\tau'} = -\frac{1}{2}e^{-\tau'/2} M_{\tau'} d\tau' + e^{-\tau'/2} dM_{\tau'} = -\frac{1}{2}V_{\tau'} d\tau' + dB_{\tau'}, \quad (\text{D.7})$$

so V is an Ornstein–Uhlenbeck process with mean-reversion $\frac{1}{2}$ and unit diffusion, whose invariant law is $\mathcal{N}(0, 1/(2 \cdot \frac{1}{2})) = \mathcal{N}(0, 1)$. The exit set $E = \{|v| > z_\alpha\}$ has invariant measure

$$\int_{\{|v| > z_\alpha\}} \varphi(v) dv = 2\Phi(-z_\alpha) = \alpha > 0, \quad (\text{D.8})$$

φ the standard normal density and the last equality by $z_\alpha = \Phi^{-1}(1 - \alpha/2)$. The OU process is a positive-recurrent (ergodic) diffusion (Meyn and Tweedie, 2009), so by the ergodic theorem

$$\frac{1}{T} \int_0^T \mathbf{1}\{|V_s| > z_\alpha\} ds \xrightarrow{\text{a.s.}} \alpha > 0, \quad (\text{D.9})$$

hence $\{\tau' : |V_{\tau'}| > z_\alpha\}$ is unbounded a.s.; by continuity of V the first entry $H := \inf\{\tau' > \log a_0 : |V_{\tau'}| > z_\alpha\}$ is finite a.s., and therefore the holding time $\tau = e^H < \infty$ a.s.

Crash channel, correct held model ($\hat{\lambda} = \lambda^*$). The centered count $M_a := N_a - \lambda^*a$ has, at integer ages, increments $\xi_i := (N_i - N_{i-1}) - \lambda^*$ that are i.i.d., mean zero, variance λ^* . By the Hartman–Wintner law of the iterated logarithm (Hartman and Wintner, 1941) applied to the integer-age walk M_n , and interpolated over unit intervals—within which the increment is at most the unit-interval count, whose maximum to age a grows as $O(\log a)$ a.s. by Borel–Cantelli against the Poisson tail, and $\log a = o(\sqrt{a \log \log a})$ —one has

$$\limsup_{a \rightarrow \infty} \frac{N_a - \lambda^*a}{\sqrt{2\lambda^*a \log \log a}} = 1, \quad \liminf_{a \rightarrow \infty} \frac{N_a - \lambda^*a}{\sqrt{2\lambda^*a \log \log a}} = -1 \quad \text{a.s.} \quad (\text{D.10})$$

By Lemma D.1 the band half-width about the held mean is of order $z_\alpha \sqrt{\lambda^*a} = O(\sqrt{a})$, and since

$$\frac{\sqrt{2\lambda^*a \log \log a}}{z_\alpha \sqrt{\lambda^*a}} = \frac{\sqrt{2 \log \log a}}{z_\alpha} \rightarrow \infty \quad (a \rightarrow \infty), \quad (\text{D.11})$$

$N_a - \lambda^*a$ exceeds the ceiling deviation $+z_\alpha \sqrt{\lambda^*a}$ for a sequence of ages tending to infinity a.s. by (D.10); the first such age is finite, so $\tau < \infty$ a.s.

Incorrect held model (held rate $\neq$ truth). The two paragraphs above dispose of a *correct* held model in either channel. An incorrect held model exits as well: its standardized statistic carries a divergent drift. In the crash channel, holding a rate $\hat{\lambda} \neq \lambda^*$, $N_a - \hat{\lambda}a = (\lambda^* - \hat{\lambda})a + M_a$; by the law of large numbers $N_a/a \rightarrow \lambda^*$ a.s., so $M_a = o(a)$ a.s., and with the band half-width $z_\alpha \sqrt{\hat{\lambda}a} = O(\sqrt{a}) = o(a)$ (Lemma D.1) the signed distance from the count to the relevant edge is

$$N_a - (\hat{\lambda}a \pm z_\alpha \sqrt{\hat{\lambda}a}) = (\lambda^* - \hat{\lambda})a + M_a \mp z_\alpha \sqrt{\hat{\lambda}a} = (\lambda^* - \hat{\lambda})a + o(a) \xrightarrow{\text{a.s.}} \pm \infty$$

according as $\hat{\lambda} \leq \lambda^*$, so the count crosses the ceiling (if $\hat{\lambda} < \lambda^*$) or the floor (if $\hat{\lambda} > \lambda^*$) at a finite age a.s. In the trend channel, holding $\hat{\mu} \neq \mu^*$, $R_a = \sqrt{a}(\mu^* -$

$\hat{\mu})/\sigma + W_a/\sqrt{a}$ with $|W_a|/\sqrt{a} \leq \sqrt{2\log\log a}(1+o(1)) = o(\sqrt{a})$ a.s. by the law of the iterated logarithm, so $R_a = \sqrt{a}(\mu^* - \hat{\mu})/\sigma(1+o(1)) \rightarrow \pm\infty$ and crosses the fixed band $\pm z_\alpha$ at a finite age a.s.

So *every* held model exits a.s.: a correct rate by the Ornstein–Uhlenbeck/law-of-the-iterated-logarithm recurrence above, an incorrect rate by this divergent drift—and this is established *without* reference to which model is in force at any particular revision, so it invokes neither consolidation nor the existence of later revisions.

Iteration and non-explosion. Each revision time σ_k is a stopping time, and the fundamental's independent increments make it strong Markov at σ_k ; with the record restarted at adoption (Lemma A.6) the regime adopted at σ_k monitors a fresh record, and the conditional law of the holding time $\tau_k = \sigma_{k+1} - \sigma_k$ given $\mathcal{F}_{\sigma_k}$ depends only on the post-revision held model $\hat{\theta}_{\sigma_k}$. Every regime exits a.s. in finite time—a correct held model by the Ornstein–Uhlenbeck/law-of-the-iterated-logarithm arguments above, an incorrect one by the divergent-drift argument above—so each $\tau_k < \infty$ a.s., and by induction $\sigma_k < \infty$ for every finite k . Each τ_k is also strictly positive (Lemma A.7): the trend channel carries a probation onset $a_0 > 0$, so $\tau_k \geq a_0$; the crash channel monitors the singleton band $\{0\}$ at small ages, so an exit there waits for the first disaster.

By Lemma D.2—whose lower-tail floor (Lemma A.7) and finite menu are exactly the strict positivity and restart structure just established—the revision times do not accumulate: $\sigma_k \uparrow \infty$ a.s. With each $\tau_k < \infty$ a.s. this gives infinitely many a.s.-finite regimes, so the model is abandoned infinitely often, yielding (i) and (ii) directly, without conditioning on the future. Part (a) then composes: it supplies an a.s. finite time T° after which every revision re-selects θ° , and since $\sigma_k \uparrow \infty$ there is a finite K with $\sigma_K \geq T^\circ$, so every regime adopted from σ_K on begins with θ° and is abandoned again—a correct-on-average model revised forever. This proves (i), (ii), and the post-

consolidation statement. □

D.4 The holding-time law

Lemma D.3 (Holding-time law). *Partition $[0, t]$ as $0 = t_0 < t_1 < \dots < t_{J-1} < t_J = t$, where $t_1, \dots, t_{J-1}$ are the ages in $(0, t)$ at which the Sterne band (7) changes (finitely many) and $t_J = t$ is the terminal endpoint. In the disputed-crash-rate channel, the model-retention survival is the finite matrix product*

$$\mathbb{P}(\tau > t) = e_0^\top \left(\prod_{j=1}^J e^{Q_j s_j} P_j \right) \mathbf{1}, \quad (\text{D.12})$$

one factor per interval $(t_{j-1}, t_j]$, taken left to right in increasing j , where: $s_j = t_j - t_{j-1}$ is the interval length; Q_j is the sub-generator on the band $[l_j, u_j]$ in force on the open interval (t_{j-1}, t_j) — P_j imposing the right-continuous band at t_j —of the disaster count killed on leaving that band, with entries

$$(Q_j)_{n,n} = -\lambda^*, \quad (Q_j)_{n,n+1} = \lambda^* \quad (n < u_j), \quad l_j \leq n \leq u_j, \quad (\text{D.13})$$

the up-move from u_j being the killing; P_j ($j < J$) is the transfer matrix embedding the surviving states of band j into band $j+1$ at the true band change t_j , and $P_J = I$ at the terminal endpoint $t_J = t$ (replaced by the transfer matrix in the coincidental case that a band change falls exactly at t); e_0 places the count at its adoption value and $\mathbf{1}$ sums over surviving states. The expression is exact—no approximation—and finite at every age, with $(e^{Q_j s})_{n,n'} = e^{-\lambda^ s} (\lambda^* s)^{n'-n} / (n' - n)!$ for $l_j \leq n \leq n' \leq u_j$.*

Proof of Lemma D.3. Fix t and partition $[0, t]$ as $0 = t_0 < t_1 < \dots < t_{J-1} < t_J = t$, with $t_1, \dots, t_{J-1}$ the band-step dates of Lemma D.1 in $(0, t)$ (finite in number) and $t_J = t$ the terminal endpoint. On $(t_{j-1}, t_j]$ the band is the constant interval $[l_j, u_j]$.

The disaster count N , killed on leaving $[l_j, u_j]$, is a pure-birth chain on $\{l_j, \dots, u_j\}$: from $n < u_j$ it moves to $n+1$ at rate λ^* ; from u_j the next disaster is the killing (upper exit). This is an age-inhomogeneous phase-type construction—a finite matrix-product whose state space changes at the band-step ages, not a time-homogeneous phase-type with a fixed generator and exponential tail (Neuts, 1981). Its sub-generator is (D.13). The probability of surviving the interval, transporting the count’s distribution from t_{j-1} to t_j , is the killed semigroup $e^{Q_j s_j}$, $s_j = t_j - t_{j-1}$. At the step date t_j the band changes to $[l_{j+1}, u_{j+1}]$ and the surviving states embed into the new state space through the transfer matrix P_j (the identity on retained states, zero on states the new floor or ceiling excludes—a lower exit at the step). By this convention $e^{Q_j s_j}$ carries the count over the half-open $(t_{j-1}, t_j]$ under the old band and P_j imposes the new band *at* t_j , so the deterministic floor-step lower exit is counted as an exit at t_j and the product below returns the post-atom survival $\mathbb{P}(\tau > t_j)$; the left-limit $\mathbb{P}(\tau > t_j^-)$ omits P_j and exceeds it by that atom’s mass. (Endpoint convention, explicitly: the product returns the right-continuous, post-atom survival $\mathbb{P}(\tau > t_j)$; the pre-step law is its left limit—the same right-continuity as Lemma D.1.) Composing from $j = 1$ to J , starting from e_0 (the count at its adoption value) and summing over surviving states with $\mathbf{1}$,

$$\mathbb{P}(\tau > t) = e_0^\top \left(\prod_{j=1}^J e^{Q_j s_j} P_j \right) \mathbf{1}, \quad (\text{D.14})$$

with $s_j = t_j - t_{j-1}$, the transfer P_j ($j < J$) at the true band change t_j , and $P_J = I$ at the terminal endpoint $t_J = t$ (the transfer matrix in the coincidental case of a band change exactly at t). This is (D.12). For the entries of $e^{Q_j s}$, the matrix Q_j is upper bidiagonal with $-\lambda^*$ on the diagonal and λ^* on the superdiagonal, so $\Phi(s) := e^{Q_j s}$

solves $\Phi'(s) = Q_j \Phi(s)$, $\Phi(0) = I$, i.e. for $l_j \leq n \leq n' \leq u_j$,

$$\frac{d}{ds} \Phi_{n,n'}(s) = -\lambda^* \Phi_{n,n'}(s) + \lambda^* \Phi_{n+1,n'}(s). \quad (\text{D.15})$$

The candidate $\Phi_{n,n'}(s) = e^{-\lambda^* s} (\lambda^* s)^{n'-n} / (n' - n)!$ satisfies (D.15): with $k = n' - n$,

$$\frac{d}{ds} \left[e^{-\lambda^* s} \frac{(\lambda^* s)^k}{k!} \right] = -\lambda^* e^{-\lambda^* s} \frac{(\lambda^* s)^k}{k!} + \lambda^* e^{-\lambda^* s} \frac{(\lambda^* s)^{k-1}}{(k-1)!} = -\lambda^* \Phi_{n,n'}(s) + \lambda^* \Phi_{n+1,n'}(s), \quad (\text{D.16})$$

since $\Phi_{n+1,n'}$ corresponds to $k-1$ (with the conventions $1/(k-1)! := 0$ at $k = 0$, so the diagonal entry solves $\Phi'_{n,n} = -\lambda^* \Phi_{n,n}$, and $\Phi_{n,n'} := 0$ for indices outside $[l_j, u_j]$, so the top *starting* state $n = u_j$ takes no super-diagonal inflow), and $\Phi_{n,n'}(0) = \mathbf{1}\{n = n'\}$ gives $\Phi(0) = I$. So $(e^{Q_j s})_{n,n'} = e^{-\lambda^* s} (\lambda^* s)^{n'-n} / (n' - n)!$, the Poisson probability of $n' - n$ births in time s , as claimed. The product (D.14) is finite and exact at every t . $\square$

D.5 A stochastic-process toolkit

Three lemmas isolate the boundary-crossing steps the tail and re-selection results rest on, each with checkable hypotheses. Two discrete errors of the counting channel are named where they enter: the $o(\sqrt{a})$ gap between integer Sterne endpoints and continuous square-root boundaries, and the one-count jump overshoot at an upper exit.

Lemma D.4 (Poisson exit through square-root boundaries). *Let N be a Poisson process of rate $\lambda > 0$, let $l \leq u$ be nondecreasing integer staircases with $l(a) = \lambda a - z\sqrt{\lambda a} + o(\sqrt{a})$ and $u(a) = \lambda a + z\sqrt{\lambda a} + o(\sqrt{a})$ for some $z > 0$, and let $T =$*

$\inf\{a > a_0 : N_a \notin [l(a), u(a)]\}$, with any finite-age convention on $[0, a_0]$. Then

$$\mathbb{P}(T > t) = t^{-\theta(1+o(1))}, \quad \theta = \lambda_0(-z, z), \quad (\text{D.17})$$

the principal Dirichlet eigenvalue of $\frac{1}{2}(\partial_{vv} - v\partial_v)$ on $(-z, z)$, and the monotone-envelope brackets of the proof have exactly regularly varying survival tails of that index. Only the asymptotic standardized endpoints $\mp z$ enter: any two staircase families with the same asymptotes have tails of the same index.

Proof. Center at the unit skeleton: the centered count at integer ages, $S_k := (N_k - \lambda k)/\sqrt{\lambda}$, $k \in \mathbb{N}$, is a random walk with i.i.d. mean-zero, unit-variance increments (centered Poisson variables, all moments finite, so $\mathbb{E}[S_1^2 \log^+ |S_1|] < \infty$), and by hypothesis the band endpoints are $\lambda a \mp z\sqrt{\lambda a} + o(\sqrt{a})$, so in the centered coordinate the retention region at age k is an interval with endpoints $g_{1,2}(k) = \mp z\sqrt{k} + o(\sqrt{k})$. Because N is nondecreasing and the ceiling and floor move only upward and only at scheduled ages (by hypothesis), the continuous-age exit is bracketed by integer-age skeleton events: writing u, l for the ceiling and floor,

$$\{N_k \in [l(k+1), u(k-1)] \forall k \leq n\} \subseteq \{\tau > n\} \subseteq \{N_k \in [l(k), u(k)] \forall k \leq n\}, \quad (\text{D.18})$$

the left event forbidding any crossing at real ages: on $(k, k+1]$ the count satisfies $N_s \leq N_{k+1} \leq u(k) \leq u(s)$ against the ceiling and $l(s) \leq l(k+1) \leq N_k \leq N_s$ against the floor. (Before the probation onset the events coincide by convention; a finite-age modification and the restriction to integer t shift constants only, survival being monotone in t .) The centered bracketing boundaries carry the staircase-against-drift oscillation and need not be monotone, so bracket once more by monotone envelopes—for the centered ceiling, $\bar{g}_2(k) = \max_{j \leq k} g_2(j)$ and $\underline{g}_2(k) = \inf_{j \geq k} g_2(j)$, and symmetrically

for the floor—each monotone with the unchanged asymptote $\mp z\sqrt{k}$, the resulting survival events nested by inclusion around those of (D.18). Each outer problem then satisfies hypotheses (H) of Greenwood and Perkins (1983): i.i.d. mean-zero, unit-variance increments with $\mathbb{E}[X^2 \log^+ X] < \infty$; boundaries $f_1 \leq 0 \leq f_2$ with f_1 eventually non-increasing, f_2 eventually nondecreasing, and $f_i(k)/\sqrt{k} \rightarrow c_i$ finite (their case (H_a))—the envelopes are monotone outright, and clipping the finitely many early ages at which the sign condition could fail is a finite-age modification of the kind already noted. Theorem 5(b) there gives $\mathbb{P}(T > n) = n^{-\lambda_0(c_1, c_2)} \pi(n)$ with π slowly varying, and their Proposition 2 identifies $-\lambda_0(c_1, c_2)$ as the largest eigenvalue of $\frac{1}{2}(\partial_{xx} - x \partial_x)$ on (c_1, c_2) with Dirichlet conditions at finite endpoints; see Novikov (1982) for one-sided nonlinear boundaries by martingale methods and Denisov et al. (2025) for the surrounding modern literature. Both brackets have $(c_1, c_2) = (-z, z)$, so both survival probabilities are regularly varying with the one index $\lambda_0(-z, z)$; the inner and outer skeleton survivals are squeezed between them, giving (D.17). The one-count overshoot at an upper exit affects the exit position, not the survival event, and does not enter. The hypotheses of case (H_a) consume only the asymptotic standardized endpoints, which proves the final claim. $\square$

Lemma D.5 (Crossing at the detection scale, crash channel). *Let N have rate λ^* and let the held rate be $\hat{\lambda} \neq \lambda^*$ with detection scale $a^* = z_\alpha^2 \hat{\lambda} / (\lambda^* - \hat{\lambda})^2$. Throughout, the regime is local misspecification— $|\lambda^* - \hat{\lambda}| \rightarrow 0$, equivalently $a^* \rightarrow \infty$, at fixed λ^* —and every $o(1)$ and limit statement is taken in that regime. Write $D(a) := (\lambda^* - \hat{\lambda}) \sqrt{a/\hat{\lambda}}$ for the standardized drift—so $|D(a)| = z_\alpha \sqrt{a/a^*}$ —and $W_a := (N_a - \lambda^* a) / \sqrt{\hat{\lambda} a}$ for the standardized fluctuation, and let E be the no-false-alarm event that W does not exit $(-z_\alpha, z_\alpha)$ before the first exit T of the record from the band of Lemma D.1. In the understated case $\hat{\lambda} < \lambda^*$, on E : (i) the first exit is the upper boundary—a floor*

exit would need $W_T \leq -z_\alpha - D(T) < -z_\alpha$; (ii) the crossing identity

$$\sqrt{T/a^\star} = \frac{z_\alpha - W_{T-} + \varepsilon_T}{z_\alpha}, \quad \varepsilon_T = o(1) + O((\hat{\lambda}T)^{-1/2}), \quad (\text{D.19})$$

holds, ε_T collecting the Sterne endpoint error and the one-count overshoot in standardized units, whence, $W_{T-} \geq -z_\alpha$ on E ,

$$T \leq 4a^\star(1 + o(1)) \quad \text{deterministically on } E; \quad (\text{D.20})$$

(iii) $\mathbb{P}(T \leq a^{\star 1-\varepsilon} \mid E) \rightarrow 0$ for every $\varepsilon > 0$, so $\log T / \log a^\star \rightarrow 1$ in probability on E . The overstated case $\hat{\lambda} > \lambda^\star$ is the mirror at the floor—a scheduled exit, no overshoot—with the same bound (D.20); there the ceiling is drift-opposed and a jump-driven upper exit remains possible on E , confined to $T = O(a^{\star 1/3})$ and of conditional probability tending to zero, so the first exit is the floor with conditional probability tending to one, not pathwise (proof, Mirror).

Proof. (i). Before T the drift is nonnegative, so a floor exit, $D(a) + W_a \leq -z_\alpha + o(1)$, forces $W_a \leq -z_\alpha - D(a) + o(1) < -z_\alpha$ off a null correction, excluded on E . (ii). At the upper exit the count equals the ceiling plus the single triggering jump, so in standardized units $D(T) + W_T = z_\alpha + \varepsilon_T$ with ε_T as stated; the triggering jump moves W by $(\hat{\lambda}T)^{-1/2}$, so $W_T = W_{T-} + O((\hat{\lambda}T)^{-1/2})$ and $D(T) = z_\alpha - W_{T-} + \varepsilon_T$, which is (D.19); $W_{T-} \geq -z_\alpha$ on E gives $D(T) \leq 2z_\alpha + \varepsilon_T$, hence (D.20). (iii). A crossing at $T \leq a^{\star 1-\varepsilon}$ forces, by (D.19), $W_{T-} \geq z_\alpha(1 - a^{\star -\varepsilon/2}) - |\varepsilon_T|$ —an approach within $\eta := z_\alpha a^{\star -\varepsilon/2}(1 + o(1))$ of the upper edge without exiting. The transfer to the diffusion is a strong approximation, not a fixed-horizon limit. The Poisson increments have exponential moments, so the Komlós–Major–Tusnády coupling applies: on a common probability space, $\sup_{u \leq a} |N_u - \lambda^\star u - B(\lambda^\star u)| = O(\log a)$ a.s. (Komlós et al., 1975,

1976; Csörgő and Révész, 1981), so the standardized coupling error is $O(\log a/\sqrt{a})$; the standardized fluctuation W carries variance $\lambda^*/\hat{\lambda} = 1 + o(1)$ in the declared regime, so the coupled Brownian term is $\sqrt{\lambda^*/\hat{\lambda}}$ times the standard one, the scaling absorbed into the boundary corrections—both negligible against the layer width η for $\varepsilon < \frac{1}{2}$, and the event $\{T \leq a^{*1-\varepsilon}\}$ shrinks in ε , so that case suffices. In log age the Brownian term is *exactly* the stationary Ornstein–Uhlenbeck ($e^{-\tau'/2}B(e^{\tau'})$, the Lamperti map), and conditioning the killed Ornstein–Uhlenbeck on not exiting is the Doob h -transform by ψ_0 —the Q -process, positive recurrent on the bounded interval (Pinsky, 1985; Collet et al., 1995, 2013); the discrete side’s conditioned position law is that of Greenwood and Perkins (1983, Theorem 5(a)), density ψ_0 against the Gaussian speed measure, vanishing linearly at the edges. Under the transform an excursion from a fixed interior layer reaches the η -layer with probability $\asymp \eta/\eta_0$ against an order-one unconditioned passage, and positive recurrence makes the expected number of such excursions over the log-horizon $(1-\varepsilon)\log a^*$ of order $\log a^*$, so by the union bound $\mathbb{P}(T \leq a^{*1-\varepsilon} \mid E) \leq C \log(a^*) a^{*-\varepsilon/2} \rightarrow 0$. *Mirror.* For $\hat{\lambda} > \lambda^*$ the drift is negative; the floor exit is scheduled (no jump, no overshoot), and the identity reads $\sqrt{T/a^*} = (z_\alpha + W_T + \varepsilon_T)/z_\alpha$ with $W_T \leq z_\alpha$ on E , giving (D.20). An *upper* exit on E is not excluded pathwise: the triggering disaster can carry W over the ceiling from $W_{T-} \geq z_\alpha + |D(T)| - (\hat{\lambda}T)^{-1/2}$, which requires $(\hat{\lambda}T)^{-1/2} > |D(T)| = z_\alpha \sqrt{T/a^*}$ and so confines such exits to $T \leq (a^*/(z_\alpha^2 \hat{\lambda}))^{1/3} = O(a^{*1/3})$ and to the drift-opposed layer of width $\eta = (\hat{\lambda}T)^{-1/2}$ below the ceiling; by the edge-layer estimate of (iii) their conditional probability is $O(\log a^* \cdot a^{*-1/6}) \rightarrow 0$. On E the first exit is therefore the scheduled floor with conditional probability tending to one, and (iii) holds at the floor by the same coupling with the reflected boundary. $\square$

Lemma D.6 (Directional re-selection at the detection scale). *In the setting of Lemma D.5,*

with a finite menu, the carried odds dominated as in Assumption 3(ii), and condition (U) holding at the exit, condition on the no-false-alarm event E . Understated, $\hat{\lambda} < \lambda^*$: the exit is the ceiling, at a disaster, and the just-ended record satisfies

$$\frac{N_T}{T} = \hat{\lambda} + z_\alpha \sqrt{\hat{\lambda}/T} (1 + o(1)) \geq \hat{\lambda} + \frac{1}{2}(\lambda^* - \hat{\lambda})(1 - o(1)), \quad (\text{D.21})$$

the inequality by (D.20). The pairwise likelihood ratio of the record between $\theta > \hat{\lambda}$ and $\hat{\lambda}$ favours θ iff N_T/T exceeds

$$\frac{\theta - \hat{\lambda}}{\log(\theta/\hat{\lambda})} = \hat{\lambda} + \frac{1}{2}(\theta - \hat{\lambda}) + O((\theta - \hat{\lambda})^2/\hat{\lambda}), \quad (\text{D.22})$$

and favours $\hat{\lambda}$ over every menu rate below it since $N_T/T > \hat{\lambda}$; so the re-selected rate satisfies $\hat{\theta}' > \hat{\lambda}$ strictly on $E \cap \{T \leq (4 - \epsilon)a^*\}$ for each $\epsilon > 0$. The boundary sliver $\{T/a^* \in [4 - \epsilon, 4(1 + o(1))]\}$ requires, by (D.19), W_{T-} within $O(\epsilon)$ of the drift-opposed edge and carries conditional mass vanishing as $\epsilon \downarrow 0$ —the conditioned position density vanishes at the edges (Greenwood and Perkins, 1983, Theorem 5(a)); an exact tie self-re-adopts by the incumbent-retention priority order. Overstated, $\hat{\lambda} > \lambda^*$: the mirror at the floor, scheduled, with $N_T/T \leq \hat{\lambda} - \frac{1}{2}(\hat{\lambda} - \lambda^*)(1 - o(1))$ and $\hat{\theta}' < \hat{\lambda}$, with conditional probability tending to one on the same event (the jump-driven upper sliver of Lemma D.5 vanishing).

Proof. The exit side is Lemma D.5(i). At the ceiling $N_T = u_\alpha(T) + 1$, and Lemma D.1's leading form gives the equality in (D.21); (D.20) gives $z_\alpha \sqrt{\hat{\lambda}/T} \geq \frac{1}{2}z_\alpha \sqrt{\hat{\lambda}/a^*} (1 - o(1)) = \frac{1}{2}(\lambda^* - \hat{\lambda})(1 - o(1))$, the inequality. For (D.22), $\log \Lambda_T^{\theta/\hat{\lambda}} = N_T \log(\theta/\hat{\lambda}) - T(\theta - \hat{\lambda})$ is positive iff N_T/T exceeds the displayed ratio, whose expansion follows from $\log(\theta/\hat{\lambda}) = \frac{\theta - \hat{\lambda}}{\hat{\lambda}} (1 - \frac{\theta - \hat{\lambda}}{2\hat{\lambda}} + O((\theta - \hat{\lambda})^2/\hat{\lambda}^2))$ inverted. On $E \cap \{T \leq (4 - \epsilon)a^*\}$, $N_T/T - \hat{\lambda} \geq \frac{z_\alpha \sqrt{\hat{\lambda}/a^*}}{2\sqrt{1 - \epsilon/4}} (1 - o(1))$, which exceeds $\frac{1}{2}(\lambda^* - \hat{\lambda}) + O((\lambda^* - \hat{\lambda})^2/\hat{\lambda})$ once

the misfit is small, so the record strictly favours over $\hat{\lambda}$ every menu rate up to $\hat{\lambda} + 2(N_T/T - \hat{\lambda})(1 + o(1))$ and favours $\hat{\lambda}$ over all rates below it; under (U) such a higher rate exists, the dominated carried odds cannot overturn the strict pairwise gaps (Assumption 3(ii)), and the argmax exceeds $\hat{\lambda}$. The boundary-sliver mass and the mirror follow as stated. $\square$

Lemma D.7 (Selection robustness). *Let $\{A'_\alpha(a)\}$ be any nested family of acceptance intervals, of held-model mass exactly $1 - \alpha$ in the continuous channel and at least $1 - \alpha$ in the counting channel, whose standardized endpoints converge to $\mp z_\alpha$, the counting endpoints nondecreasing integer staircases. Then Corollary 1(a) holds at the level of the tail exponent, $\mathbb{P}(\tau > t) = t^{-\theta(\alpha)(1+o(1))}$: the trend problem has the same principal eigenvalue—the exact power law $\sim C t^{-\theta(\alpha)}$ requiring endpoints exactly $\mp z_\alpha$, merely-convergent endpoints controlling the exponent but not the slowly varying factor—and the crash tail is $t^{-\theta(\alpha)(1+o(1))}$ by Lemma D.4, whose hypotheses consume only the asymptotic endpoints. The finite/infinite-mean threshold at $\theta(\alpha) = 1$ and the recurrence of Proposition 1 are selection-invariant; the calendar of scheduled steps, the predictable-atom dates, and the finite-age geometry of the band are selection-specific. In the symmetric limits of both channels the natural selections—highest-density, equal-tailed, central—share the endpoints $\mp z_\alpha$.*

Proof. In the trend channel the standardized band lies, from some age on, between the fixed intervals $(-z_\alpha \mp \delta, z_\alpha \pm \delta)$ for every $\delta > 0$; domain monotonicity of the killed problem sandwiches the survival between the two fixed-band problems, whose exponents converge to $\theta(\alpha)$ by the continuity of λ_0 (Greenwood and Perkins, 1983, Proposition 2), so the exponent is unchanged, and with endpoints exactly $\mp z_\alpha$ the spectral representation (D.23) is unchanged including the constant. In the crash channel the hypotheses of Lemma D.4 are the endpoint asymptotics and the monotone

staircase form, both assumed, so (D.17) gives the tail, and the mean threshold follows from the lemma's brackets as in Corollary 1(a). Recurrence in Proposition 1 uses only $\tau < \infty$ a.s. and the restart, both selection-free. The scheduled dates are the integer-attainment ages of the specific staircase and move with the selection. $\square$

D.6 Heavy tails and the detection scale

Proof of Corollary 1(a). By (D.6), holding the correct trend model, $\tau = e^H$ with $H = \inf\{\tau' > \log a_0 : |V_{\tau'}| > z_\alpha\}$ the first exit of the Ornstein–Uhlenbeck process V from $(-z_\alpha, z_\alpha)$. Monitoring begins at $\tau' = \log a_0$, so the killed-process duration is $D := H - \log a_0$, with ν_0 the law of V at the onset $\log a_0$ *restricted to the interior* $(-z_\alpha, z_\alpha)$. Under the correct model $V_{\log a_0}$ is standard normal, so the onset places mass $\alpha = \mathbb{P}(|V| > z_\alpha)$ already outside the band—those paths exit at once, $D = 0$ —and only the interior sub-probability mass of total $1 - \alpha$ feeds the positive-survival expansion below (equivalently, the Dirichlet eigenfunctions, extended by zero outside the band, integrate against ν_0 over the interior alone); the atom at $D = 0$ does not affect the $s \rightarrow \infty$ tail. The generator $\mathcal{L} = \frac{1}{2}\partial_{vv} - \frac{1}{2}v\partial_v$ on $(-z_\alpha, z_\alpha)$ with Dirichlet boundary conditions is self-adjoint with respect to the Gaussian speed (invariant) measure $m(dv) \propto e^{-v^2/2} dv$, not Lebesgue measure—a regular Sturm–Liouville operator with discrete spectrum $0 < \theta_1(\alpha) < \theta_2(\alpha) < \dots$ and a positive principal eigenfunction ψ_1 , the eigenfunctions $\{\psi_k\}$ orthonormal in $L^2(m)$. The survival of the killed process admits the eigenvalue (spectral) expansion for diffusion hitting times (Kent, 1980; Alili et al., 2005)

$$\mathbb{P}(D > s) = \sum_{k \geq 1} c_k e^{-\theta_k(\alpha)s}, \quad c_1 = \left(\int \psi_1 dm \right) \left(\int \psi_1 d\nu_0 \right) > 0, \quad (\text{D.23})$$

the duration measured from the onset, so $\mathbb{P}(D > s) \sim c_1 e^{-\theta_1(\alpha)s}$ as $s \rightarrow \infty$. Therefore, using $H = \log a_0 + D$,

$$\mathbb{P}(\tau > t) = \mathbb{P}(e^H > t) = \mathbb{P}(D > \log(t/a_0)) \sim c_1 e^{-\theta_1(\alpha) \log(t/a_0)} = (c_1 a_0^{\theta_1(\alpha)}) t^{-\theta_1(\alpha)}, \quad (\text{D.24})$$

which is (10) with $\theta(\alpha) = \theta_1(\alpha)$ and constant $C = c_1 a_0^{\theta_1(\alpha)}$ depending on the onset a_0 but not the exponent. The mean is

$$\mathbb{E}[\tau] = \mathbb{E}[e^H] = a_0 \int_0^\infty e^s (-d\mathbb{P}(D > s)) < \infty \iff \theta_1(\alpha) > 1, \quad (\text{D.25})$$

the integral converging at $s \rightarrow \infty$ exactly when the decay rate $\theta_1(\alpha)$ in (D.23) exceeds 1.

In the crash channel, holding the correct model the count has rate λ^* , and by Lemma D.1 the Sterne staircases are nondecreasing with endpoints $\lambda^* a \mp z_\alpha \sqrt{\lambda^* a} + o(\sqrt{a})$ —the integer-versus-continuous gap is exactly the $o(\sqrt{a})$ the hypotheses of Lemma D.4 absorb—so the lemma applies with $(\lambda, z) = (\lambda^*, z_\alpha)$ and (D.17) is (11), the index the same principal Dirichlet eigenvalue as the trend channel's, the operator being the same. The finite-mean threshold follows from the lemma's exactly regularly varying brackets: $\mathbb{E}[\tau] < \infty$ when $\theta(\alpha) > 1$ (the upper bracket integrable) and $\mathbb{E}[\tau] = \infty$ when $\theta(\alpha) < 1$ (the lower bracket not). The jump structure and the predictable-floor/inaccessible-ceiling split of the exit (Lemma D.1) enter the exact finite-age law (D.12) but wash out of the index.

For an incorrect held model the statistic carries a drift (Corollary 1(b)), so this heavy tail is a property of holding the correct model. $\square$

Proof of Corollary 1(b). Trend. With $\hat{\mu} \neq \mu^*$, $R_a = \sqrt{a}(\mu^* - \hat{\mu})/\sigma + W_a/\sqrt{a}$; the deterministic term reaches the band edge z_α when $\sqrt{a}|\mu^* - \hat{\mu}|/\sigma = z_\alpha$, i.e. at

$a = z_\alpha^2 \sigma^2 / (\mu^\star - \hat{\mu})^2$, the bounded-in-probability fluctuation $W_a / \sqrt{a} = O_P(1)$ being of the same order as the band half-width z_α —so $a^\star$ is the detection *scale*, not a sharp threshold, and the concentration is the conditional statement of Corollary 1(b). *Crash.* With $\hat{\lambda} \neq \lambda^\star$, $N_a - \hat{\lambda}a = (\lambda^\star - \hat{\lambda})a + M_a$ reaches the band half-width $z_\alpha \sqrt{\hat{\lambda}a}$ when $|\lambda^\star - \hat{\lambda}|a = z_\alpha \sqrt{\hat{\lambda}a}$, i.e. at $a = z_\alpha^2 \hat{\lambda} / (\lambda^\star - \hat{\lambda})^2$. In both the deterministic drift reaches the band at the $a^\star$ of (12): in the trend channel the standardized R_a has an $O(1)$ band and $O(1)$ fluctuation against a drift $\sqrt{a}(\mu^\star - \hat{\mu})/\sigma$ that grows like $\sqrt{a}$, while in the crash channel the unstandardized $N_a - \hat{\lambda}a$ has drift $(\lambda^\star - \hat{\lambda})a$ against an $O(\sqrt{a})$ martingale and band. The crossing is *shallow*, so $a^\star$ pins only the *scale*: in logarithmic age $\tau' = \log a$ the trend statistic is $\delta e^{\tau'/2} + V_{\tau'}$ with $\delta = (\mu^\star - \hat{\mu})/\sigma$ —for $\mu^\star < \hat{\mu}$ reflect the record, $X \mapsto -X$, the band being symmetric, so take $\delta > 0$ without loss—and V the stationary Ornstein–Uhlenbeck process of (D.6) (order one); the deterministic part reaches z_α at $\tau'^\star = \log a^\star$ with slope $z_\alpha/2$. *Conditional on no earlier false alarm*—no exit of the stationary fluctuation across the fixed band before the crossing—the statistic equals z_α at the upper crossing H , i.e. $\delta e^{H/2} + V_H = z_\alpha$ with $V_H \in [-z_\alpha, z_\alpha)$ on the conditioning event (V is continuous, and $V_H < z_\alpha$ since $\delta e^{H/2} > 0$), and the relation is exact, requiring no expansion and no tightness of V at the random time H :

$$e^{(H-\tau'^\star)/2} = \frac{z_\alpha - V_H}{z_\alpha} \in (0, 2] \quad \implies \quad H \leq \tau'^\star + 2 \log 2 \quad \text{deterministically on the event.} \quad (\text{D.26})$$

For the matching lower bound, a crossing at $H \leq (1 - \varepsilon)\tau'^\star$ forces, by the same exact relation, $V_H \geq z_\alpha(1 - e^{-\varepsilon\tau'^\star/2})$ —an approach within $\eta_\varepsilon := z_\alpha e^{-\varepsilon\tau'^\star/2}$ of the absorbing edge. Under the conditioned law the edge is entrance-repelling: conditioning on survival is the Doob h -transform by the principal eigenfunction ψ_0 —the positive recurrent Q -process of the killed Ornstein–Uhlenbeck on a bounded interval (Pinsky,

1985; Collet et al., 1995)—under which a passage from the fixed interior layer $z_\alpha - \eta_0$ to the η -layer carries the factor $\psi_0(z_\alpha - \eta)/\psi_0(z_\alpha - \eta_0) \asymp \eta/\eta_0$ against an order-one unconditioned passage (ψ_0 vanishes linearly at the edge), and the ψ_0 -conditioned diffusion, being positive recurrent, makes an *expected* $O(\tau'^*)$ approaches to the fixed layer over the horizon, so by the union bound

$$\mathbb{P}(H \leq (1 - \varepsilon)\tau'^* \mid \text{no earlier false alarm}) \leq C \tau'^* e^{-\varepsilon\tau'^*/2} \longrightarrow 0 \quad \text{for every } \varepsilon > 0.$$

In the crash channel the two bounds are Lemma D.5(ii)–(iii): the crossing identity (D.19) replaces the exact relation, the one-count overshoot enters only through ε_T , and the Komlós–Major–Tusnády strong approximation enters only the lower side. Hence, on the conditioning event, $H/\tau'^* \rightarrow 1$ in probability, so the holding time concentrates on the *log* scale, $\log \tau / \log a^* \rightarrow 1$ (equivalently $\tau = a^{*1+o_{\mathbb{P}}(1)}$) as the misfit $\rightarrow 0$. The multiplicative ratio τ/a^* is not claimed to converge: the upper side is the deterministic $\tau/a^* = e^{H-\tau'^*} \leq 4$, the lower side only sub-polynomial. The limit is conditional on no earlier false alarm, and this conditioning event’s probability *vanishes* as the misfit $\rightarrow 0$ —over the growing log-age interval before the deterministic crossing, the stationary fluctuation exits the fixed band with probability tending to one—so the *unconditional* holding time is governed by the fixed-band false-alarm time of the stationary fluctuation—an a.s. finite random time, tight relative to the diverging a^* but not concentrating around any deterministic shorter scale. Thus $\log \tau / \log a^* \rightarrow 1$ is the no-false-alarm statement, and a^* is the conditional detection scale. For the matched comparison, work in the local (small-misfit) regime, where $\text{KL}(\mathbb{P}^* \parallel \mathbb{P}^{\hat{\lambda}}) \approx (\lambda^* - \hat{\lambda})^2 / (2\lambda^*)$ (Poisson) and likewise in the trend channel; equal KL then fixes $|\lambda^* - \hat{\lambda}|$ to leading order, and the band coefficient orders a^* only at the next order. Away from this regime equal KL no longer pins the misfit term of (12),

and the comparison is stated only locally. $\square$

D.7 The two deaths

The general theory of this appendix is channel-free. The crash channel adds one sharp, channel-specific feature, the *two deaths* flagged in §2.5: the *direction* of a model's error fixes the *manner* in which a long-enough record refutes it. We state the result and its scope, then prove it.

Assumption 3 (Local misspecification / long record). *The crash-channel re-selection-direction result—Proposition 4—is stated for the local-misspecification regime: $\hat{\lambda}$ is close enough to λ^* that the detection scale $a^* = z_\alpha^2 \hat{\lambda} / (\lambda^* - \hat{\lambda})^2$ (Corollary 1(b)) is large; the two consequences below are statements about exits at that scale—the conditional regime of Corollary 1(b)—while fluctuation-driven exits at short records occur and are outside them. Two consequences are used. (i) Understated rate, $\hat{\lambda} < \lambda^*$: at an upper exit the count sits on the ceiling, $u_\alpha(\tau) = \hat{\lambda}\tau + z_\alpha \sqrt{\hat{\lambda}\tau} + o(\sqrt{\tau})$, so $N_\tau/\tau = \hat{\lambda} + z_\alpha \sqrt{\hat{\lambda}/\tau} + o(1/\sqrt{\tau})$; on the no-false-alarm event $\tau \leq 4a^*(1 + o(1))$ (Lemma D.5), so $N_\tau/\tau \geq \hat{\lambda} + \frac{1}{2}(\lambda^* - \hat{\lambda})(1 - o(1))$, equal to λ^* to leading order at $\tau/a^* \rightarrow 1$ ($z_\alpha \sqrt{\hat{\lambda}/a^*} = \lambda^* - \hat{\lambda}$); the post-jump count exceeds the ceiling by the single triggering disaster, a further $o(1)$ correction. Overstated rate, $\hat{\lambda} > \lambda^*$: the detection-side exit is the floor, $l_\alpha(\tau) = \hat{\lambda}\tau - z_\alpha \sqrt{\hat{\lambda}\tau} + o(\sqrt{\tau})$, with the opposite sign, $N_\tau/\tau = \hat{\lambda} - z_\alpha \sqrt{\hat{\lambda}/\tau} + o(1/\sqrt{\tau}) \leq \hat{\lambda} - \frac{1}{2}(\hat{\lambda} - \lambda^*)(1 - o(1))$, and $z_\alpha \sqrt{\hat{\lambda}/a^*} = \hat{\lambda} - \lambda^*$. (ii) The re-selection (8) is driven by the record rather than by ρ when the carried log-odds at adoption are dominated by the record's pairwise separations, $\max_{\theta \neq \theta'} |\log(\rho(\theta)/\rho(\theta'))| = o(\Delta_\tau)$ with Δ_τ the smallest pairwise log-likelihood-ratio gap the just-ended record generates among the menu rates it ranks (a uniform sufficient condition: the selection turns only on the gaps to contenders that can overtake*

the record-favoured model)—the operative scale for a local-alternatives menu, whose pairwise gaps shrink relative to τ ; for a fixed menu of separated rates the gaps are of order τ and the criterion is correspondingly weaker, as when the prior history is short or the carried ranking near-uniform—and not after consolidation, where the carried ranking already favours θ° by a margin growing with the accumulated history, the re-selection is ρ -driven and self-re-adopts θ° (condition (U) fails); the directional map below, a statement about detectably-misspecified held rates, then does not apply, and we restrict it to the $o(\tau)$ case. For gross misspecification the record is short, the carried prior can decide the re-selected rate, and the map is not guaranteed. The exit-side and timing statements hold conditional on no earlier false alarm (Lemma D.5) and need neither consequence (i) nor (ii); only the re-selection direction additionally uses them.

One condition governs whether an exit’s *direction* translates into a strictly different re-selected rate. Call the held rate *detectably misspecified* on its just-ended record when it is not the front-runner of that record,

$$\hat{\lambda} \neq \arg \max_{\theta \in \Theta} L_\theta(X_{(\sigma, \sigma + \tau]}) \rho(\theta). \quad (\text{U})$$

Condition (U) holds whenever the model is detectably misspecified—an *environment-specific* property of the held rate against its record and the menu, not a behavioral parameter—and fails ($\hat{\lambda}' = \hat{\lambda}$, self-re-adoption) when the held rate remains the record’s MAP front-runner: not only when it equals the empirical rate, but whenever the empirical rate falls in its likelihood cell or the carried ranking keeps it on top, as for a consolidated or correct model, whose record’s empirical rate $\rightarrow \lambda^*$ lies in $\hat{\lambda}$ ’s likelihood cell— $\hat{\lambda}$ the menu rate nearest λ^* in Kullback–Leibler, equal to λ^* when the truth is on the menu—and favours $\hat{\lambda}$ exponentially over any other menu rate.

Proposition 4 (Two deaths, disputed-crash-rate channel). *In the disputed-crash-rate channel, conditional on the record surviving long enough to detect the misfit—to an age of order the detection scale a^* of Corollary 1(b), without an earlier exit—the exit side is fixed by the error direction—with conditional probability tending to one in the local-misspecification limit, on the no-earlier-false-alarm event of Lemma D.5—and the re-selection direction by the record’s empirical rate. A held model that understates the disaster rate, $\hat{\lambda} < \lambda^*$, is then carried to its ceiling by the rising count and dies by an upper exit—a disaster carrying the count above the ceiling, a totally inaccessible stopping time of the count’s filtration—and re-selects (8) a rate weakly above $\hat{\lambda}$, strictly above under (U)—the exit record’s empirical rate sitting at least halfway from $\hat{\lambda}$ toward λ^* (Lemma D.6; Assumption 3). A held model that overstates it, $\hat{\lambda} > \lambda^*$, is carried below its floor and dies by a lower exit—the scheduled step-up of the floor in a quiet spell, a predictable stopping time—and re-selects a rate weakly below $\hat{\lambda}$, strictly below under (U)—at least halfway down toward λ^* (Lemma D.6). The conditioning is essential, not cosmetic: unconditionally, before the misfit is detected, an early chance fluctuation can trigger either exit, and Corollary 1(b) shows these false alarms dominate the holding time as the misfit shrinks—so the directional map is the conditional-on-detection statement, not an unconditional claim that a misspecified model usually dies on its drift-aligned side.*

Proof of Proposition 4. Decompose the count relative to the held mean,

$$N_a - \hat{\lambda}a = (\lambda^* - \hat{\lambda})a + M_a, \quad M_a := N_a - \lambda^*a, \quad (\text{D.27})$$

with M_a a mean-zero martingale, $M_a = O_{\mathbb{P}}(\sqrt{a})$ and $M_a = o(a)$ a.s. by (D.10). By Lemma D.1 the ceiling and floor sit at $\pm z_{\alpha} \sqrt{\hat{\lambda}a} = O(\sqrt{a})$ about the held mean.

Understatement, $\hat{\lambda} < \lambda^$.* The drift $(\lambda^* - \hat{\lambda})a > 0$ in (D.27) grows linearly and

dominates both $M_a = o(a)$ and the $O(\sqrt{a})$ ceiling, so for all large a , $N_a - \hat{\lambda}a$ exceeds the ceiling: the count exceeds the ceiling at a finite age almost surely, so the ceiling is the boundary the drift reaches. Whether this *upper* exit is the *first* exit is settled below; the displayed crossing shows only that the ceiling is reached a.s. eventually. An upper exit occurs at a jump of N , i.e. at a disaster; the jump times of a Poisson process are totally inaccessible stopping times of its natural filtration, admitting no announcing sequence (Dellacherie and Meyer, 1982). On re-selection (8) after a conditional upper exit the just-ended record has empirical rate at least halfway from $\hat{\lambda}$ to λ^* ((D.21), via the deterministic crossing bound (D.20); Assumption 3), so its likelihood favours a *strictly higher* menu rate when condition (U) holds, while a $\hat{\lambda}$ that remains the menu rate nearest the record self-re-adopts ((U) fails), $\hat{\lambda}' = \hat{\lambda}$. The drift-opposite event is a lower exit—the rising floor catching a stalled count. Since N_a is non-decreasing while the floor $\hat{\lambda}a - z_\alpha \sqrt{\hat{\lambda}a}$ rises only at rate $\hat{\lambda} < \lambda^*$, the gap

$$N_{a'} - (\hat{\lambda}a' - z_\alpha \sqrt{\hat{\lambda}a'}) = (\lambda^* - \hat{\lambda})a' + M_{a'} + z_\alpha \sqrt{\hat{\lambda}a'} = (\lambda^* - \hat{\lambda})a' + o(a') \rightarrow +\infty \quad \text{a.s.}$$

grows: the count outpaces the floor, and by (D.10) it also exceeds the ceiling at a finite age a.s. A lower exit can nonetheless come *first*: the floor catches the count whenever an early downward fluctuation of M of order the band half-width $z_\alpha \sqrt{\hat{\lambda}a}$ arrives before the drift has separated the count from the floor, an event of positive probability that does not vanish with age. Which exit is first is thus a competition between the drift-driven ceiling crossing (near a^*) and an early lower false alarm, and we do *not* claim the upper exit wins unconditionally. On the contrary, Corollary 1(b) shows that in the local-misspecification regime (a^* large) the fixed-band false-alarm time *dominates* the detection scale, so unconditionally the first exit is typically a false alarm, which can fall on either band. What survives is the *conditional* statement,

and it is Lemma D.6: conditional on *no earlier false alarm*—the same regime as Corollary 1(b)—the first exit is the ceiling, pathwise on the conditioning event (a floor exit would require the fluctuation itself to cross the band, Lemma D.5(i)); it arrives by (D.20) no later than $4a^*(1 + o(1))$; and the just-ended record’s empirical rate (D.21) sits at least halfway from $\hat{\lambda}$ to λ^* , so the likelihood re-selects a strictly higher menu rate under (U) and self-re-adopts otherwise. No survival-conditioned concentration estimate enters: the conditioning excludes the drift-opposed boundary pathwise, and the re-selection direction follows from the deterministic crossing bound alone.

Overstatement, $\hat{\lambda} > \lambda^*$. Symmetrically the drift $(\lambda^* - \hat{\lambda})a < 0$ carries the count behind the band centre, the floor gap $N_{a'} - (\hat{\lambda}a' - z_\alpha \sqrt{\hat{\lambda}a'}) = -(\hat{\lambda} - \lambda^*)a' + o(a') \rightarrow -\infty$ a.s.; the unconstrained count falls behind the floor at a finite age almost surely—an eventual, counterfactual crossing, not yet the spell’s first exit; when the lower exit *is* first, it occurs at a scheduled band-step date t_j where the floor steps up past the count. That date is deterministic and the event $\{N_{t_j^-} < l_{j+1}\}$ is $\mathcal{F}_{t_j^-}$ -measurable, so the lower exit is a predictable stopping time. Re-selection then returns, under (U), a rate *strictly lower* than $\hat{\lambda}$ —the exit record’s empirical rate at least halfway down toward λ^* (Lemma D.6, *Overstated*)—and self-re-adopts otherwise. Symmetrically the drift-opposite event is an upper exit—a disaster surge carrying the lagging count over the ceiling before the floor catches it—of positive, non-vanishing probability, so again the first exit is not unconditionally the lower one; conditional on no earlier false alarm the mirror of Lemma D.6 applies: the exit is the floor—an upper exit would require the fluctuation itself to cross the band—scheduled, arriving by (D.20) no later than $4a^*(1 + o(1))$, with exit empirical rate at least halfway from $\hat{\lambda}$ down to λ^* , so the likelihood re-selects a strictly *lower* menu rate under (U); the next boundary touched is the floor, with conditional probability tending to one on the conditioning

event (Lemma D.5, *Mirror*). The error-direction map—understate $\Rightarrow$ upper (inaccessible) exit $\Rightarrow$ revise toward fear, by surprise; overstate $\Rightarrow$ lower (predictable) exit $\Rightarrow$ revise toward calm, on schedule—is therefore the *conditional-on-detection* statement: it names the boundary the misfit-driven count reaches once a long enough record has detected the error, not an unconditional claim about which exit comes first, since before detection a chance false alarm can refute either type on either side (Corollary 1(b)). $\square$

E Aggregate belief dynamics: proofs

This appendix proves the results of Section 3. Throughout, the common path of X is fixed, $\Psi_t : \Xi \rightarrow \mathcal{S}$ is the flow it induces on belief states, and $m_t = \Psi_t \# G$.

Proof of Proposition 2. For fixed path, the band half-width $z_{\alpha(\xi)}$ depends measurably on ξ , and the successive exit times are first passage times of the common record against the ξ -dependent band, each a measurable functional of (ξ, path) ; hence Ψ_t is G -measurable and $m_t = \Psi_t \# G$ is a well-defined element of $\mathcal{P}(\mathcal{S})$. The identity $\langle \varphi, m_t \rangle = \int_{\Xi} \varphi(\Psi_t(\xi)) G(d\xi)$ is the change-of-variables formula for a pushforward and uses no averaging over independent agent-level noise: the only randomness is the common path, held fixed. For the realized-versus-ex-ante distinction, fix an age a and let $\mathbf{1}_{\text{surv}}^a(\xi)$ indicate that type ξ 's current holding spell has survived from adoption to age a —the event whose probability (C.2) computes—so that, conditional on the spell's initialization—adoption date, held model, carried ranking, the state from which (C.2) is computed—its expectation is the single-type survival law from that state, $\mathbb{E}[\mathbf{1}_{\text{surv}}^a(\xi)] = S_{\xi}(a)$ —the type- ξ survival along the realized path from adoption, carrying the adoption date through the path segment it spans; the unconditional mixture is adoption-aligned, over G and the initialization law, not a calendar-time snapshot.

The realized population survivor fraction to age a , $\int_{\Xi} \mathbf{1}_{\text{surv}}^a(\xi) G(d\xi)$, is a functional of the path, and by Fubini $\mathbb{E} \int_{\Xi} \mathbf{1}_{\text{surv}}^a(\xi) G(d\xi) = \int_{\Xi} \mathbb{E}[\mathbf{1}_{\text{surv}}^a(\xi)] G(d\xi) = \int_{\Xi} S_{\xi}(a) G(d\xi)$, the G -mixture of survival laws—so the realized fraction equals the ex-ante mixture only in expectation, and neither equals the single-type $S(a)$. This survivor-to-age object is distinct from the *instantaneous* non-revision fraction that enters m_t in the statement—the G -mass of types not revising in the instant at t , a separate pathwise functional with its own mean—which is precisely why the cross-section $m_t = \Psi_t \# G$ is an instantaneous pushforward of G and not a survival mixture. $\square$

Proof of Corollary 2. We derive (14) as a direct pathwise balance, not from an extended generator. Fix the realized common path. Each type ξ then has a deterministic trajectory: along the common path the record $R_t = T_{a_t}$ is a deterministic function of ξ and t —the same data windowed from ξ 's adoption date—the held model $\hat{\theta}$ is constant between revisions, and the age rises at unit rate, so $\Psi_t(\xi) = (\hat{\theta}_t(\xi), \alpha(\xi), a_t(\xi))$ is determined— $\dot{\alpha} = 0$, the tolerance a frozen coordinate—and $m_t = \Psi_t \# G$ (Proposition 2). The record's between-revision motion is a diffusion in the trend channel and piecewise-constant in the crash channel; conditioning on the common path makes it a fixed sample path either way, so no autonomous deterministic flow—and no Davis piecewise-deterministic structure—is invoked or needed. A revision is the first exit of that record from the band $A_{\alpha}(a)$ of (4): the first time the trajectory $R_t(\xi)$ leaves the ordinary range, at which instant $\Psi_t(\xi)$ jumps by the re-selection-and-restart map J of (8) (a new held model, the record reset to age 0, testing resuming at the probation onset a_0). No transversality of the crossing is required: the first-exit time is well defined for each type along the path, the diffusion's local recrossings of the boundary notwithstanding, since the record resets at the first exit and the old band is no longer monitored.

Differentiate $\langle \varphi, m_t \rangle = \int_{\Xi} \varphi(\Psi_t(\xi)) G(d\xi)$ in t along the path. Between its own exits $\frac{d}{dt} \varphi(\Psi_t(\xi)) = \partial_a \varphi(\Psi_t(\xi))$, since $\hat{\theta}$ is fixed and $\dot{a} = 1$; at an exit φ jumps by $(\varphi \circ J - \varphi)(\Psi_{t-}(\xi))$. Integrate over ξ (Fubini; φ bounded, G a probability measure) and collect the exits into the *crossing flux* Φ_t —the measure on pre-jump states that accumulates, by time t , the G -mass of types whose record has first reached the band edge, marked by where it crosses—so that the transport term is $\langle \partial_a \varphi, m_t \rangle dt$ and the boundary-crossing term is the Stieltjes integral $\int (\varphi \circ J - \varphi) d\Phi_t$ against Φ_t , which is locally finite—its cumulative mass on $[0, t]$ is the G -average number of crossings by t , finite on compacts by a *channel-specific* argument rather than a single floor. In the trend channel the deterministic probation floor $\tau_k \geq a_0 > 0$ (Lemma A.7) caps each type’s revision count on $[0, t]$ at t/a_0 , so the G -average is finite. In the crash channel, where $a_0 = 0$ and no deterministic floor holds, crossings occur only at disasters and at scheduled floor-steps: disasters arrive as a Poisson process of rate λ^* , a.s. finitely many on $[0, t]$, and the scheduled exits are uniformly spaced from below across the continuum of types: within any spell the Sterne floor first reaches 1 only at an age $a_1(\hat{\lambda}, \alpha) > 0$ that grows as the band widens and shrinks as the held rate rises, so $a_1 \geq a_1(\lambda_K, \bar{\alpha}) > 0$ uniformly, with λ_K the largest menu rate and $\bar{\alpha} := \sup\{\alpha(\xi) : \xi \in \text{supp } G\}$, below 1 by the non-degeneracy clause of Assumption 2. A spell ends at a disaster or at a scheduled exit, and any scheduled exit falls at an age of at least $a_1(\lambda_K, \bar{\alpha})$; so between consecutive disasters, a gap of length g holds at most $g/a_1(\lambda_K, \bar{\alpha}) + 1$ spells, and every type’s crossing count on $[0, t]$ obeys the single integrable bound $2N_t + t/a_1(\lambda_K, \bar{\alpha}) + 1$, the G -average finite on compacts—local finiteness inherited from the driving Poisson process and the uniform first-step floor, not from a holding-time floor. In neither channel is the count bounded by 1: under recurrence it grows with the population-average number of revisions. The hazard (9) is the *ex-ante* rate of these boundary hits (Appendix C), the expectation over paths of the flux, not a pathwise intensity; Φ_t itself

is the realized, pathwise crossing flux, consistent with the pathwise m_t and not with $\mathbb{E}[m_t]$ for independently-revising agents. Differentiating the fixed-population integral $\int_{\Xi} \varphi(\Psi_t(\xi)) G(d\xi)$ closes with these two terms *alone*—transport and flux—because G is a fixed probability measure over a closed population: no mass enters or leaves and every type carries equal weight. The remaining two channels are not produced by this differentiation; each requires changing what is differentiated. Turnover enters once the population is *not* closed: it is a genuine rate- p birth–death jump appended to the balance—at rate p a G -uniform fraction is replaced and reinjected at the type’s newborn state $s_o(\xi)$ —each agent reborn with her own tolerance, the newborn cross-section the pushforward $m_o = s_o \# G$ —contributing $p(\langle \varphi, m_o \rangle - \langle \varphi, m_t \rangle) dt$. Selection enters once the weighting is *not* head-count. Write the wealth-weighted cross-section as the tilt $m_t^w(ds) = \omega_t(s) m_t(ds) / \langle \omega_t, m_t \rangle$ by relative wealth share ω_t , with $d\omega_t / \omega_t = r_t dt$ and $r_t(s)$ the relative return (net of the wealth-weighted average). Differentiating $\langle \varphi, m_t^w \rangle = \langle \varphi \omega_t, m_t \rangle / \langle \omega_t, m_t \rangle$ by the quotient and product rules, the weight motion contributes

$$d\Sigma_t(\varphi) = (\langle \varphi r_t \rangle_{m_t^w} - \langle \varphi \rangle_{m_t^w} \langle r_t \rangle_{m_t^w}) dt = \text{Cov}_{m_t^w}(\varphi, r_t) dt, \quad (\text{E.1})$$

the replicator covariance expressed in share space, identically zero under the head-count weighting taken as the baseline here. Collecting: the head-count average carries the first three contributions, which is (14); and between common disasters, where the weight motion is continuous, the quotient rule above returns the same integrals under the tilt and adds $d\Sigma_t$, which is (15)—at a common disaster weights and beliefs jump together, and the joint atom is the (17) object, outside these displays. The terms of (14): the ageing drift $\langle \partial_a \varphi, m_t \rangle dt$ (transport); the boundary-crossing contribution $\int (\varphi \circ J - \varphi) d\Phi_t$ (flux; J the re-adoption map, which resets the *age* coordinate to

0—the probation onset a_0 is the distinct object gating when testing resumes on the fresh record), where Φ_t resolves into three parts, none absolutely continuous under the baseline assumptions: a *continuous* part from the heterogeneously-timed crossings of the continuous statistic, singular-continuous in general and continuous under atomless G ; *predictable atoms* at scheduled lower exits when a positive cohort shares a reset age (the floor step-ups of Lemma D.1); and an *inaccessible atom* at a common disaster, where one jump pushes every count up at once and all types with ceiling below the new count exit together, $\Delta\Phi_u = b_u$ (the inaccessible upper exits of Proposition 4); the turnover term $p(\langle\varphi, m_o\rangle - \langle\varphi, m_t\rangle) dt$; and, when the measure is wealth-weighted with $d\pi/\pi$ the relative performance, the product-rule term $d\Sigma_t(\varphi) = \text{Cov}_{m_t^w}(\varphi, \text{relative return}) dt$ (selection). The predictable atoms require a *tolerance-independent* rank-swap date: only floor-steps at such dates synchronize a positive cohort across an atomless G (the shared reset ages of Lemma D.1), while floor movements tied to *tolerance-dependent* cumulative thresholds disperse, a rank-swap step contributing an atom only over the positive-mass set of tolerances it re-orders. Predictable atoms are thus a crash-channel feature—absent in the trend channel, where dispersed crossing times leave no shared reset age—but not generic across all floor movements.

Ex-ante crossing rate. Let $\mu_t \in \mathcal{P}(\bar{\mathcal{S}})$ denote the law at t of the full enlarged state $\bar{s} = (\hat{\theta}, T_a, a, \rho)$ of Theorem 1(iii), and $m_t = \Pi_{\#}\mu_t$ its $(\hat{\theta}, a)$ -marginal under $\Pi : \bar{s} \mapsto (\hat{\theta}, a)$. The ex-ante expected crossing rate is channel-specific. In the *crash channel* the exit carries the genuine state-dependent rate χ of (C.3), so the expected crossing rate is $\langle\chi, \mu_t\rangle = \int_{\bar{\mathcal{S}}} \chi(\hat{\theta}, T_a, a; \alpha) d\mu_t(\bar{s})$, the tolerance α a coordinate of the type carried by the state; because $\chi(\hat{\theta}, x, a; \alpha) = \lambda^* \mathbf{1}\{x = u_\alpha(a)\}$ depends on the record only through the event $\{T_a = u_\alpha(a)\}$ at the type's *own* ceiling, the inner integral over the record fibre evaluates, type by type, to $\lambda^* \nu_a(\{u_\alpha(a)\} \mid \hat{\theta}, \alpha, a) = h(\hat{\theta}, \alpha, a)$ and

it pushes down to the single-spell hazard of each tolerance type, $\langle \chi, \mu_t \rangle = \langle h, m_t \rangle$ with h carrying the $(\hat{\theta}, \alpha, a)$ arguments—no cross-type averaging, the fibres being tolerance-specific—the pushdown well defined because the cross-section m_t of (13) carries the tolerance coordinate; and $\langle \chi, \mu_t \rangle$ is the *inaccessible upper-exit* component of the crash-channel crossing rate, the scheduled floor-step exits being the predictable-atom part of Φ_t , outside χ . In the *trend channel* the exit is a predictable boundary hit carrying no *pathwise* rate χ , so the pathwise pushdown above has no trend analogue; the ex-ante marginal hazard h of (9) remains well defined from the holding-time law, but the object that closes the flux term of (14) is the boundary flux Φ_t itself, not $\langle h, m_t \rangle$. $\square$

Proof of Corollary 3. At a common-surprise date u the only mass that moves is the crossing mass b_u , carried by the post-revision map J from its pre-jump location to $J\#b_u$, the interior of m being unaffected: $m_u - m_{u-} = J\#b_u - b_u$. For a linear index,

$$\Delta \langle \varphi, m_u \rangle = \int \varphi d(J\#b_u) - \int \varphi db_u = \int (\varphi \circ J - \varphi) db_u, \quad (\text{E.2})$$

which is (16). For a nonlinear aggregate $\Gamma(\langle \varphi, m \rangle)$ the jump is the finite increment $\Gamma(\langle \varphi, m_u \rangle) - \Gamma(\langle \varphi, m_{u-} \rangle)$; this is not $\Gamma'(\langle \varphi, m_{u-} \rangle) \Delta \langle \varphi, m_u \rangle$ because the jump in $\langle \varphi, m \rangle$ is of finite size, not infinitesimal, so the first-order expansion does not close. $\square$

F Stakes-invariance: the record, not the payoff

A final property of the rule bears on *why* people revise, and it is the model's structural restriction—though the observation behind it is not ours. Because the trigger is the record and not the payoff, the revision rate does not depend on the stakes of being wrong. This is exactly the feature that would distinguish an inference rule from a cost

or rational-inattention one (Sims, 2003; Baley and Turén, 2025)—but the distinction is subtle and not ours to claim: Cohen et al. (2018) show that, to first order for a small switching cost, the optimal band of inaction *coincides* with a two-sided test, so cost and inference generate the same static band, and only their *dynamic* response to the stakes tells them apart. We describe that response precisely, as a property our rule has by construction; the constant-hazard sticky-information benchmark (Mankiw and Reis, 2002) shares it, so the contrast is only with state- and cost-dependent accounts. The proof is in Appendix G.

Definition 2 (Hybrid revision hazard). *To separate the two accounts empirically, write a reduced-form revision propensity for investor i at record age a as an additive nesting of a baseline clock, a surprise term, and a stakes term,*

$$\lambda_i(a) = \lambda_0(a) + h_S(\bar{\varepsilon}_i(a)) + h_C(\Delta V_i(a) - K_i), \quad (\text{F.1})$$

with λ_0 a Calvo baseline (Calvo, 1983); h_S increasing in the running surprise score $\bar{\varepsilon}_i(a)$, a normalization of the record’s position in the held model’s tail that rises as the record grows less likely (so the test of Definition 1 fires when $\bar{\varepsilon}_i$ crosses the size- α level); and h_C increasing in the net value of revising $\Delta V_i(a) - K_i$ over its observed support $\mathcal{V}$. The two accounts are poles: at the cost pole h_C is strictly increasing over $\mathcal{V}$; at the inference pole $h_C \equiv 0$ on $\mathcal{V}$ and revision is governed by h_S alone, recovering the first-exit trigger of Definition 1 in the limit where h_S concentrates on the band edge and $\lambda_0 \rightarrow 0$. Three hazard objects must be kept apart here: the internal trigger—the first-exit compensator, in general a measure with predictable atoms; the marginal age hazard $h(a)$ of (C.3), its average over surviving record positions; and the observed model-change hazard, the trigger thinned by self-re-adoption (Lemma G.1). The nesting (F.1) specifies the intensity component of the internal trigger conditional on the

surprise score $\bar{\varepsilon}_i(a)$ —a smoothed stand-in for the first-exit trigger, not the marginal $h(a)$ and not the observed hazard; averaging over the surviving score distribution yields the marginal, and thinning by π the observed.

The hybrid makes inference and cost two poles of one nested, falsifiable model, and the inference pole—the limit $h_C \equiv 0$ on $\mathcal{V}$, $\lambda_0 \rightarrow 0$ of Definition 1—satisfies the following invariance exactly.

Proposition 5 (Stakes-invariance). *Under the pure-inference rule of Definition 1, the revision time τ of (5) is a functional of the test, the menu, and the record alone, invariant to any strictly monotone transformation of the payoff stakes that holds the data-generating environment fixed; it does not depend on ΔV . The reduced-form counterpart is an identifying restriction on (F.1): conditional on the running surprise $\bar{\varepsilon}_i(a)$ and the rest of the inference state, the inference-pole hazard does not load on stake proxies (h_C flat on $\mathcal{V}$).*

Stakes-invariance holds because the trigger contains no payoff argument: a believer abandons a model when its record looks too unlikely under it, regardless of how much the belief is worth—the content of the inference channel, and exactly what a cost or strategic account denies.

The separating contrast is sharper than surprise dependence versus none. The cost hazard $h_C(\Delta V_i(a) - K_i)$ can itself co-move with surprise, since the net value of revising depends on the gap between held and optimal belief, which the surprise score also tracks; and a merely *constant* h_C is absorbed into λ_0 and not separately identified. The clean discriminant is *stake sensitivity conditional on surprise*: holding the running surprise fixed, the cost-pole hazard still responds to stake variation over $\mathcal{V}$ while the inference-pole hazard does not. The two make opposite predictions about one covariate: a cost account (Baley and Turén, 2025; Sims, 2003) predicts revision

grows more frequent as the stakes of being wrong, or the distance to the consensus, rise; the inference account predicts that, holding surprise fixed, neither matters.

One observational subtlety connects the two literatures. A panel records a *change in the reported model*, not every internal revision: an investor who abandons a model and re-adopts it registers no observed change, so consecutive holding times compound into one observed spell. This self-re-adoption thinning is the inference-side account of the reported-versus-internal-belief wedge the cost literature emphasizes (Baley and Turén, 2025)—observed stickiness overstates internal stickiness in both accounts—so the two agree on why reported revisions are sparse and disagree only on what moves their timing. Crucially, the thinning is itself a stakes-free functional of the inference state (Lemma G.1, Appendix G): the *observed* hazard is stakes-invariant too, so the panel prediction of §2.2 holds for observed revisions, not just the internal trigger.

Testing this is a genuine identification problem, and not one this paper solves. The restriction holds conditional on the hidden inference state (R_t, a_t, ρ_t) (Lemma G.1), so a naive regression of the observed hazard on a stakes proxy is confounded by the unobserved ρ_t . Two routes break the confound—exogenous stake variation at a fixed inference state, or explicit hidden-state estimation—and reduce it to a conditional moment: with $\Delta_{it} \in \{0, 1\}$ an observed model change, $\bar{\varepsilon}_{it}$ the observed-spell surprise score, and s_{it} a stake measure, surprise-testing imposes

$$\mathbb{E}[\Delta_{it} \mid \bar{\varepsilon}_{it}, s_{it}, R_t, a_t, \rho_t] = \mathbb{E}[\Delta_{it} \mid \bar{\varepsilon}_{it}, R_t, a_t, \rho_t] : \quad (\text{F.2})$$

the model-change probability does not load on the stake once surprise and the inference state are held fixed. A cost- or attention-governed threshold violates it with a strictly positive loading—higher stakes, faster revision at the same surprise. An agent revises because her model has failed a test, not because failing has grown expensive.

G The hybrid hazard: proof

Proof of Proposition 5. By (5), $\tau = \inf\{a > a_0 : T_a \notin A_\alpha(a)\}$. The record T_a is a functional of the observed data path; the acceptance region $A_\alpha(a) = \{f_a^{\hat{\theta}} \geq c_\alpha(a)\}$ of (4) is the held model’s predictive highest-density region at size α (Lemma A.3), a functional of the held model’s law. Let g be a strictly monotone transformation of the payoff stakes that leaves the data-generating environment fixed. Then g changes ΔV_i but leaves both the data path—hence T_a —and the held model’s predictive law—hence $A_\alpha(a)$ —unchanged. Therefore the event $\{T_a \notin A_\alpha(a)\}$ is unchanged for every a , and so is its first time τ . This proves invariance. For the converse, at the cost pole h_C is strictly increasing over $\mathcal{V}$, so a stake transformation that moves ΔV_i over $\mathcal{V}$ strictly alters h_C , hence the revision intensity (F.1) and the distribution of the revision time. A merely constant h_C shifts only the level and is absorbed into the baseline λ_0 , leaving timing stake-insensitive and unidentified separately from λ_0 ; the identified, testable content is therefore the *loading* of the hazard on stake variation over its observed support $\mathcal{V}$, not the presence of a cost term. $\square$

Lemma G.1 (Observable stakes-invariance). *Under the pure-inference pole ($h_C \equiv 0$ on $\mathcal{V}$), the internal revision hazard is $\lambda_0(a) + h_S(R_t)$, a functional of age and record alone. Let the observed revision process count only model-changing revisions—self-re-adoptions, which leave the held model unchanged, registering no observed change. Then the observed revision process is the self-re-adoption thinning of the internal trigger at the compensator level: its compensator is $\pi d\Lambda^{\text{int}}$, where Λ^{int} is the internal trigger’s compensator—an ordinary intensity where one exists, as in the nesting (F.1), and in general the hazard measure of (C.1), with its predictable trend-boundary and scheduled floor-step atoms—the thinning probability π evaluated at the trigger’s mark, the stopped record (the post-disaster count at a crash upper exit; the record at the*

atom under the right-continuous convention), acting through the mark kernel of the compensator (Jacod, 1975) while the intensity itself is predictable from the pre-jump state—so that where the intensity exists,

$$\lambda_t^{\text{obs}} = [\lambda_0(a_t) + h_S(R_t)] \pi(R_t, a_t, \rho_t), \quad \pi(R_t, a_t, \rho_t) = \mathbb{P}(\hat{\theta}' \neq \hat{\theta} \mid R_t, a_t, \rho_t), \quad (\text{G.1})$$

with π the model-change probability—a function of the record, the regime age, and the carried ranking ρ_t , which fixes the front-runner. Both factors are functionals of the inference state (R_t, a_t, ρ_t) and the primitives—menu, prior ρ_0 , recording environment, baseline λ_0 —alone, free of the stakes $\Delta V_i(a) - K_i$; the age a_t enters because the baseline clock $\lambda_0(a)$ is age-dependent and the model-change probability depends on the record over its regime length. Hence the observed hazard is stakes-invariant conditional on the inference state. The record alone does not determine the held model, so π is not a function of R_t only; but ρ_t is itself a stakes-free functional of the history. The restriction is thus that, conditional on the full inference state (R_t, a_t, ρ_t) , no stakes covariate loads on the observed hazard—a structural statement, not an observational one: self-re-adoptions reset the internal record without a reported change, so a reported spell need not reveal (R_t, a_t, ρ_t) , and the empirical identification is taken up in Appendix F.

Proof of Lemma G.1. Under $h_C \equiv 0$ on $\mathcal{V}$ the internal hazard is $\lambda_0(a) + h_S(R_t)$ (Definition 2), a functional of age and record alone (Sections 2–2.4); no stakes enter the baseline clock, the acceptance band, or the surprise statistic. Condition on a trigger at the stopped pair (R, a) —the record summary and its regime length, the pair the stopped likelihood consults. The re-selection (8) returns $\hat{\theta}' = \arg \max_{\theta} L_{\theta}(R, a) \rho(\theta)$, a function of the stopped record and the carried ranking alone, so the event $\{\hat{\theta}' \neq \hat{\theta}\}$ —condition (U)—is $\sigma(R, a, \rho)$ -measurable and carries no stakes argument; its probabil-

ity given the inference state is $\pi(R, a, \rho)$. The observed point process is the internal hazard thinned by this event, so its intensity is the product (G.1). Each factor is a stakes-free functional of the inference state and the primitives—the internal hazard $\lambda_0 + h_S$ by Proposition 5, π by the stakes-free re-selection rule (8)—so the product is stakes-free. Conditioning on the full inference state (R, a, ρ) —all stakes-free functionals of the history—the observed hazard does not depend on the stakes. $\square$

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